# Auditing Correlated Failures in Frozen-Feature Pretrained-Encoder Pools for Medical Segmentation

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## Abstract

Medical segmentation ensembles are credited with complementary failure coverage when independently trained members make different mistakes. We audit this assumption for frozen and lightly-adapted pretrained-encoder pipelines, where independently trained decoders can share the same backbone, pretraining bundle, and recipe. Across five primary task rows (Kvasir polyp, ACDC LV, BraTS foreground, RIGA Cup, RIGA Disc), our 11-bundle $\times$ 4-decoder-seed pool yields per-case Dice correlation gaps of $\Delta = 0.261$–$0.638$ between same-bundle and cross-bundle pairs after a Dice $\geq 0.30$ functional floor; all paired case and hierarchical bootstrap intervals exclude zero. The result is stable under floor sweeps, ICC(A,1), leave-one-out, subject aggregation, quality-controlled pair regression, and cross-fitted item-difficulty residualisation. Diagnostic probes (pixel error, joint-failure lift, same-architecture-family ViT, same-family cross-checkpoint, same-backbone) locate the dependence structure but do not causally separate architecture, pretraining, and recipe. The claim is scoped to 2D frozen/light-adaptation encoder pools and retrospective benchmark audits. We release per-case Dice traces, summary JSONs, scorer code, hashes, and metadata as the audit artifact.

## 1 Introduction

Ensembles are often used in safety-relevant medical segmentation because different members may fail on different cases. Aggregate Dice can tell us whether members are accurate, but it does not by itself show whether additional members provide additional failure evidence on the same cases. This paper audits that missing quantity for frozen or lightly adapted pretrained-encoder segmentation pools: the aligned case-level correlation of segmentation quality across pool members.

The medical-imaging supply chain now reuses a small set of foundation-model encoders and conventional pretrained backbones: DINOv2, CLIP/BiomedCLIP, SAM/MedSAM, ConvNeXt, EfficientNet, ResNet, and domain-specific pretrainings such as RETFound. A pool of "different" segmenters can therefore share backbone family, pretraining bundle, input statistics, and training recipe. This creates an evaluation gap: aggregate Dice and common uncertainty summaries usually do not directly report aligned case-level co-failure, which is the evidence needed for independent failure-coverage claims. The intuition that near-duplicate pipelines should share errors is not new; the missing piece is a released, case-aligned audit that quantifies how large the dependence is, tests whether it survives simple alternative explanations, and states which evaluation claims the evidence can and cannot support.

**Primary contribution.** We contribute a reusable E&D audit protocol for fixed segmentation pools: compute per-case Dice traces, define a prespecified primary-table same/release-recipe group and cross-bundle reference, average symmetrically over decoder seeds, and report paired bootstrap intervals for the same/cross correlation gap. The protocol asks a concrete reporting question: before

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a pool is credited with complementary failure coverage, have same/release-recipe and cross-bundle co-failure correlations been measured on aligned cases? Out of scope: prompt-conditioned MedSAM, full nnU-Net pipelines, clinical outcomes, demographic fairness, and SOTA segmentation performance.

**Artifact and diagnostic contribution.** We release the trace-level files and scorer code needed to recompute the primary audit statistics. Additional diagnostics test estimator sensitivity, subject-level clustering, item difficulty, pixel error overlap, same-family cross-checkpoint structure, same-backbone perturbations, adaptation and full-pipeline boundaries, and selected scope checks; these diagnostics are not used for causal decomposition. The reusable object contributed by this paper is therefore an audit protocol plus a trace-level evidence bundle for measuring whether a fixed pool of segmentation models supplies independent failure evidence on aligned cases.

## 2 Related work

**Relation to prior work and scope of novelty.** Prior work has established that correlated errors and monoculture can matter. Our narrower contribution is to make this issue auditable for a concrete medical segmentation setting by releasing aligned per-case traces, same/cross estimators, and a primary/diagnostic evidence hierarchy. Our risk framing follows algorithmic-monoculture and foundation-model ecosystem work [Kleinberg and Raghavan, 2021, Bommasani et al., 2021, 2022, Toups et al., 2023]. The correlation gap we measure has modelling antecedents: trial-by-trial error consistency asks whether systems fail on the same inputs [Geirhos et al., 2020], error correlations arise when deployed models share algorithms, datasets, or foundation models [Li et al., 2025], shared training trajectories and weight-space basins can constrain checkpoint diversity [Fort et al., 2019, Wortsman et al., 2022], transfer/representation work studies what is reused across pretrained models [Neyshabur et al., 2020, Gontijo-Lopes et al., 2022], pretrained-model ensemble work treats pretraining as a possible diversity source [Mustafa et al., 2020], same-architecture models can align strongly across initializations [Kornblith et al., 2019], and predictive-diversity pathologies can limit deep-ensemble gains [Abe et al., 2024]. Recent LLM monoculture and null-model papers are relevant because they show that agreement/correlation claims depend on the choice of baseline and unit of analysis; our paper applies that lesson to case-level medical segmentation traces [Jiang et al., 2025, Kim et al., 2025a, Gorecki and Hardt, 2025, Goel et al., 2025, Jo et al., 2026]. Pathology-FM and precision-oncology studies provide adjacent evidence about center robustness, benchmark behavior, and complementary domain-specialised features [de Jong et al., 2025, Campanella et al., 2025, Luo et al., 2025, Neidlinger et al., 2025]. The distinction is threefold: our unit is an aligned segmentation Dice trace rather than a classification decision or deployment outcome, our artifact releases the trace-level evidence needed to recompute the audit rather than only a summary statistic, and the paper separates primary same-checkpoint/release-recipe evidence from mechanism-adjacent diagnostics. Our contribution is not the general claim that correlated errors exist. It is a released medical-segmentation audit of frozen-feature pretrained-encoder pools that distinguishes primary evidence from diagnostics and reports where same-checkpoint/recipe, same-architecture-family, and broader cross-family correlations appear under a shared protocol.

**Medical segmentation, pretrained/foundation-model benchmarks, and ensemble baselines.** nnU-Net, deep ensembles, MC dropout, adjacent uncertainty or parameter-efficient ensembles, and hyperparameter ensembles are usually evaluated through Dice, calibration, or uncertainty quality [Isensee et al., 2021, 2024, Lakshminarayanan et al., 2017, Gal and Ghahramani, 2016, Mehrtash et al., 2020, Jungo and Reyes, 2019, Mühlematter et al., 2026, Gu et al., 2024]. Classical ensemble-diversity metrics motivate this audit, but our reporting target is the case-aligned correlation of task-performance traces because that statistic directly tests redundant failure evidence on the evaluated benchmark [Kuncheva and Whitaker, 2003, Ganaie et al., 2022, Wood et al., 2023]. Medical-segmentation diversity and failure-detection benchmarks ask whether a method can detect segmentation failures; our audit asks whether multiple pool members fail on the same cases, making the questions complementary rather than substitutable [Georgescu et al., 2023, Zenk et al., 2025]. Recent medical-segmentation benchmarks and robustness-specification work evaluate or discuss generic-vision probes, SAM medical-image evaluations, standardized segmentation datasets, promptable SAM adaptations, universal 3D/medical segmenters, task-specific robustness requirements, transferability estimation, and VFM segmentation benchmarking [Baharoon et al., 2023, Huang et al., 2024, Kuş and Aydin, 2024, Kirillov et al., 2023, Ma et al., 2024, Cheng et al., 2023, Archit et al., 2025, Zhu et al., 2024, Ma et al., 2025, Gu et al., 2025, He et al., 2025, Kim et al., 2025b, Xian et al., 2025, Yang et al., 2023, Kerssies et al., 2024]. These benchmarks are complementary to our audit: they ask which model or foundation model performs well, whereas we ask whether multiple members of a pool provide non-redundant failure evidence on the same cases. Promptable and 3D medical foundation models are deployment alternatives rather than baselines for this co-failure audit.

# 3 Formal setup and benchmark estimands

**Pool and estimands.** Let $\mathcal{F}=\{F_1,\ldots,F_K\}$ be the operational pretrained-encoder bundles used by the Table 1 scorer. In the primary 11-bundle source pool, a bundle is the concrete encoder / pretraining / input-statistic wrapper used for frozen-feature extraction. Same-bundle decoder seeds form most of the main same group, and DINOv2-B$\times$DINOv2-S is the only recurring cross-checkpoint member of that group because both variants share the same upstream DINOv2 release and pretraining recipe. Other broad-family groupings (for example OpenAI CLIP sizes, ResNet sizes, and expanded DINOv2/ConvNeXt variants) are not folded into the Table 1 primary estimator; they are separated in the same-family cross-checkpoint diagnostic in Appendix A30. BiomedCLIP (ViT-B/16, biomedical image-text pretraining) is treated as a different bundle from DINOv2 despite being a ViT. A model trace is a tuple $(F_k,\sigma)$ (bundle, decoder seed) producing per-case Dice $D_{(k,\sigma)}(i,t)$. Appendix A29 and Appendix A30 report diagnostics that separate primary product bundles, same-architecture-family ViT / different-pretraining, and broader same-family cross-checkpoint readouts; these clarify that a "cross" column can still contain broad architectural structure. For a pair of models $a\neq b$ we define $r_{a,b}(t)=\mathrm{Pearson}_i[D_a(i,t),D_b(i,t)]$ (and a McGraw–Wong ICC(A,1) variant $\iota_{a,b}$ [McGraw and Wong, 1996], chosen to penalise per-bundle Dice offsets that Pearson absorbs). The primary-table same/cross means at task $t$ are the expectations of $r_{a,b}$ over pairs in the same/release-recipe group (same bundle, plus DINOv2-B/S) and the remaining cross-bundle reference under this scorer. We define $\Delta(t)=\bar{r}_{\mathrm{same/release}}(t)-\bar{r}_{\mathrm{cross}}(t)$ as the co-failure gap and refer to it as the *primary same/cross gap*, $\Delta$, throughout. Secondary estimands are pixel-level error-mask Dice, computed the same way over binarised error masks, and joint-failure *lift* over independence $L_{\mathrm{mono}}(\tau)=\Pr[\text{all fail}\mid\text{mono pool}]/\prod_m \Pr[\mathrm{fail}_m\mid\tau]$.

**Terminology used by the audit.** The primary claim uses only the primary-table same/release-recipe group and the cross-bundle reference; broader family rows are diagnostics.

| Term | Meaning in this paper | Primary claim? |
|---|---|---|
| Bundle | Concrete encoder checkpoint plus wrapper, pretraining, input normalisation, and feature-extraction path | Yes |
| Model trace | Bundle plus decoder seed and aligned per-case Dice vector | Yes |
| Primary-table same/release-recipe group | Same-bundle seed pairs plus the prespecified DINOv2-B/S same-release cross-checkpoint pair | Yes |
| Primary-table cross reference | Remaining primary-pool pairs under the primary grouping | Yes |
| Same-family cross-checkpoint | Broader DINOv2, CLIP, ConvNeXt, and ResNet diagnostic grid | Diagnostic only |
| Same-architecture ViT cross-pretraining | ViT-B diagnostic varying pretraining source | Diagnostic only |

**Symmetric multi-seed averaging.** For each bundle pair we enumerate all $|\Sigma|^2=16$ seed combinations (4 per model) rather than seed-matched pairs, removing the one-seed-per-bundle noise that inflates ad-hoc gap estimates. Same-bundle averaging uses $\binom{|\Sigma|}{2}=6$ same-bundle seed pairs per $F_k$ plus the full $|\Sigma|^2$ on DINOv2-B$\times$DINOv2-S; the cross group uses all remaining primary-table bundle pairs.

**CI and scope.** A 3-level nested hierarchical bootstrap (primary groups $\to$ seeds $\to$ cases, $B=2000$) yields 95% CIs as the 2.5th/97.5th percentiles of bootstrap-sampled $\bar{r}_{\mathrm{same}},\bar{r}_{\mathrm{cross}},\Delta$. Because the number of encoder-bundle clusters is small, neither interval should be read as a population-level guarantee over all possible pretrained encoders. We report both intervals as finite-audit uncertainty summaries: the case bootstrap asks about new cases conditional on the evaluated pool, while the hierarchical bootstrap also perturbs the finite group/seed surface. The estimands assume a shared-decoder, shared-task pool and measure *observed* correlation within such pools; a CNN random-init diagnostic (Appendix A31.2) shows positive same-vs-cross structure without pretrained weights, so we report the pretrained-bundle gap descriptively rather than as a causal isolation of pretrained-weight effects.

**Primary-group structural caveat.** The primary rows retain task-specific functional pools of 11/10/10/9/11 bundles after the Dice floor. In these pools, the Table 1 "same" column is dominated by same-bundle, different-seed decoder-head pairs, with DINOv2-B×DINOv2-S providing the only recurring same-family cross-checkpoint pair when both variants are functional. Thus the high same-group $\bar{r}$ values in Table 1 ($0.834$–$0.959$) are best read as same-frozen-checkpoint/recipe-dominated dependence readouts, not a broad sample of many same-family cross-checkpoint comparisons. Section 6.1 and Appendix A29 report a 5-variant same-architecture-family ViT / different-pretraining diagnostic (DINOv2-B, BiomedCLIP, MAE-B, DeiT-B, CLIP-B) that measures cross-pretraining correlation under a shared broad architecture family and is the closest "within-architecture-family, cross-pretraining" control in this benchmark.

# 4 Experimental design

**Datasets & targets.** The primary rows are summarized below. The splits are recorded in the released JSON files' `test_ids` fields; subject-level sensitivity for ACDC and BraTS is in Appendix A11.

**Primary task rows and evaluation units.** The audit uses benchmark-derived targets and aligned per-case Dice traces, not clinical deployment cohorts.

| Task row | Raw source | Target / split | Unit | Main caveat |
|---|---|---|---|---|
| Kvasir polyp | Kvasir-SEG [Jha et al., 2020] | Loader-defined 880/120 image split | image, $N=120$ | No patient/video IDs available; no patient-level duplicate control claimed. |
| ACDC LV | ACDC [Bernard et al., 2018] | LV-positive ED/ES slices from public patients 001–100, fold 0 | slice, $N=361$; 20 patients | Slice row; patient-level sensitivity reported in Appendix A11. |
| BraTS foreground | BraTS 2024 GLI [BraTS 2024 Organizers, 2024, de Verdier et al., 2024] | 2D FLAIR `seg>0`; top-10 eligible slices per held-out subject, NumPy seed 42 | slice, $N=3{,}228$; 324 subjects | Includes resection-cavity label; not the official 3D BraTS WT challenge target. |
| RIGA Cup/Disc | RIGA+ [Almazroa et al., 2018, Almazroa, 2018] | BinRushed+MESSIDOR train, Magrabia source test | fundus, $N=95$ | Deterministic hashed alignment IDs retained only for split alignment. |

**Encoder bundles.** *Generic pretrained-encoder bundles* (11, mixing foundation-model encoders and conventional pretrained backbones): DINOv2-B/S, BiomedCLIP, CLIP ViT-L/14, CLIP ViT-B/16, MAE ViT-B/16, DeiT ViT-B/16, ConvNeXt-T, EfficientNet-B0, ResNet-50/18. Encoder names refer to the exact released wrapper/checkpoint paths in the artifact; in particular, CLIP-L/14 is the documented OpenCLIP dense-token wrapper rather than a claim of official OpenAI inference parity. *Medical-domain encoder probe:* MedSAM vision-encoder-only (SAM ViT-B; prompt encoder and mask decoder discarded) is reported as a secondary boundary probe on RIGA/Kvasir/ACDC. RETFound support is documented as a gated-checkpoint path, but RETFound cells are not reported because the gated checkpoint was unavailable in the release environment.

**Protocol.** The primary protocol resizes images and masks to $224\times224$ with nearest-neighbor mask resizing, applies encoder-specific upstream normalisation (ImageNet-style statistics for DINOv2/MAE/DeiT/torchvision CNNs and OpenAI CLIP statistics for CLIP/BiomedCLIP), freezes the encoder, and trains a $1\times1$-conv decoder with Adam $10^{-3}$ for 30 epochs, batch 16, BCE loss on the binary primary rows, no early stopping, no validation-based model selection, and fixed sigmoid threshold 0.5 at inference. Table 1 starts from the 11-bundle generic source pool where traces are available and applies the estimator's Dice$\geq$0.30 retrospective functional floor to remove non-segmenting, near-constant traces: Kvasir retains 11 bundles, ACDC 10, BraTS foreground 10, RIGA Cup 9, and RIGA Disc 11. This test-trace-based analytic convention defines the reported functional pool; it is not a training, deployment, or pre-registered hypothesis threshold. The floor systematically reduces cross-bundle correlation (near-constant outputs would be near-zero correlation with everything), so the post-floor pool biases the gap upward by construction; the released floor sweep (Appendix A5) shows that primary-row signs do not flip across $\{0, 0.20, 0.25, 0.30, 0.35, 0.40\}$, but the magnitudes are floor-conditional. The MedSAM sensitivity is a secondary encoder-only boundary probe, not part of the primary pool or a prompt-conditioned MedSAM evaluation.

# 5 Main result: same-frozen-checkpoint pools show higher co-failure correlation

The broad same-family question is separated from the seed-dominated primary readout before interpreting Table 1. Appendix A30 recomputes an expanded diagnostic that isolates distinct checkpoints within broad upstream families from same-bundle seed pairs; the same-family cross-checkpoint minus diagnostic-cross gap remains positive on all five rows (0.080–0.267) but is far below the same-bundle seed floor. Table 1 is therefore a finite-pool same-frozen-checkpoint/release-recipe audit of the failure-evidence assumption, not a claim that all broad same-family checkpoints are equally redundant.

Table 1: **Primary finite-pool co-failure audit.** For each filtered functional pool, we report average pairwise Pearson correlation of per-case Dice traces for the primary same/release-recipe group and the cross-bundle reference. $K$ is the retained bundle count after the Dice$\geq$0.30 functional floor; $M$ is the number of retained model traces; pair counts are same/cross model-trace pairs. $\Delta_{0.30}$ is conditional on that retained functional pool, while $\Delta_{\text{no floor}}$ recomputes the same scorer without the floor. CIs are 95% bootstrap intervals for $\Delta_{0.30} = \rho_{\text{same/release}} - \rho_{\text{cross}}$. ACDC and BraTS are slice-level rows nested within 20 ACDC patients and 324 BraTS subjects respectively; subject-aggregated gaps are ACDC $\Delta = 0.262$ [0.151, 0.413] and BraTS $\Delta = 0.214$ [0.182, 0.251] (Appendix A11). The BraTS row uses the 11-bundle source pool for the 2D FLAIR `seg>0` derivative; ResNet-18 is below the Dice floor, leaving 10 retained bundles. Primary ACDC pool retains CLIP-L/14 (passes the Dice $\geq$ 0.30 floor); a secondary 8-FM ACDC diagnostic with a different split (Appendix A13) reports CLIP-L/14 collapsed at Dice 0.007 on that split.

| Task row | Unit $N$ | $K$ | $M$ | pairs S/C | $\rho_{\text{same/release}}$ | $\rho_{\text{cross}}$ | $\Delta_{0.30}$ | $\Delta_{\text{no floor}}$ | $\Delta$ case CI* | $\Delta$ hier CI† |
|---|---|---|---|---|---|---|---|---|---|---|
| Kvasir polyp | image 120 | 11 | 44 | 82/864 | 0.958 | 0.697 | 0.261 | 0.261 | [0.211,0.320] | [0.174,0.378] |
| ACDC LV | slice 361 (20 pts) | 10 | 40 | 76/704 | 0.959 | 0.639 | 0.321 | 0.355 | [0.289,0.355] | [0.211,0.435] |
| BraTS foreground | slice 3,228 (324 subj) | 10 | 40 | 76/704 | 0.938 | 0.623 | 0.315 | 0.362 | [0.302,0.329] | [0.224,0.425] |
| RIGA Cup | fundus 95 | 9 | 36 | 70/560 | 0.834 | 0.361 | 0.473 | 0.529 | [0.395,0.556] | [0.280,0.780] |
| RIGA Disc | fundus 95 | 11 | 44 | 82/864 | 0.870 | 0.232 | 0.638 | 0.638 | [0.569,0.687] | [0.475,0.831] |

* Case-level paired bootstrap ($B$=2,000), conditional on the functional pool shown. † Hierarchical bootstrap over primary grouping and seed clusters; it is a finite-pool interval conditional on the evaluated panel, and all displayed hierarchical CIs exclude zero in this audit.

Across four imaging domains and five task rows, the filtered functional pools contain 36–44 model traces. Within these finite pools, the same/release-recipe group has higher average per-case Dice correlation than the cross-bundle reference on every row ($\Delta_{0.30} = 0.261$–$0.638$). The CIs in Table 1 summarize uncertainty for this finite audit, not a population-level claim over future encoders, datasets, or recipes. The retrospective functional floor is visible in the table: no-floor gaps are also positive (Kvasir 0.261, ACDC 0.355, BraTS 0.362, RIGA Cup 0.529, RIGA Disc 0.638), and the calibration-selected disjoint-evaluation gaps in Table 2 remain positive. For the nested slice rows, subject aggregation attenuates the magnitude to ACDC 0.262 and BraTS 0.214 while preserving the sign; Kvasir remains image-level evidence because patient/video IDs are unavailable. Table 2 summarizes four sign-stability diagnostics: a calibration-selected functional pool evaluated on disjoint cases, a binary failure-event readout at the empirical $P_{33}$ threshold, quality-controlled pair regression, and distinct-checkpoint same-family decomposition.

**Estimator robustness.** A sensitivity summary (Appendix A5) recomputes the primary Pearson gap under a functional-floor sweep, an ICC(A,1) alternative, leave-one-bundle/task-out summaries, and a random family-label permutation stress test. Analytic-floor checks show the gap remains positive at no-floor and stricter-floor settings where the pool is non-empty. Event and ICC checks show the same sign after binarising failures at the empirical $P_{33}$ threshold and after replacing Pearson with ICC(A,1). Quality and residualisation checks show that simple marginal-quality controls and held-out item-difficulty residualisation do not absorb the same/cross separation. The random-label permutation is retained only as an appendix stress test because family labels are not exchangeable sampling units; it is not used as a main-text significance claim or population-level $p$-value.

**Spatial and lift readouts.** Figure 1 is not used as primary evidence for the Table 1 claim. It illustrates why continuous case-level correlation is worth auditing: in selected cases, same-bundle seeds can produce spatially overlapping false positives and false negatives, and secondary aggregate pixel/lift diagnostics show the same qualitative dependence pattern. In the aggregate diagnostics, same-family pairwise error-mask Dice is 0.80–0.87 versus 0.32–0.40 cross-family on RIGA

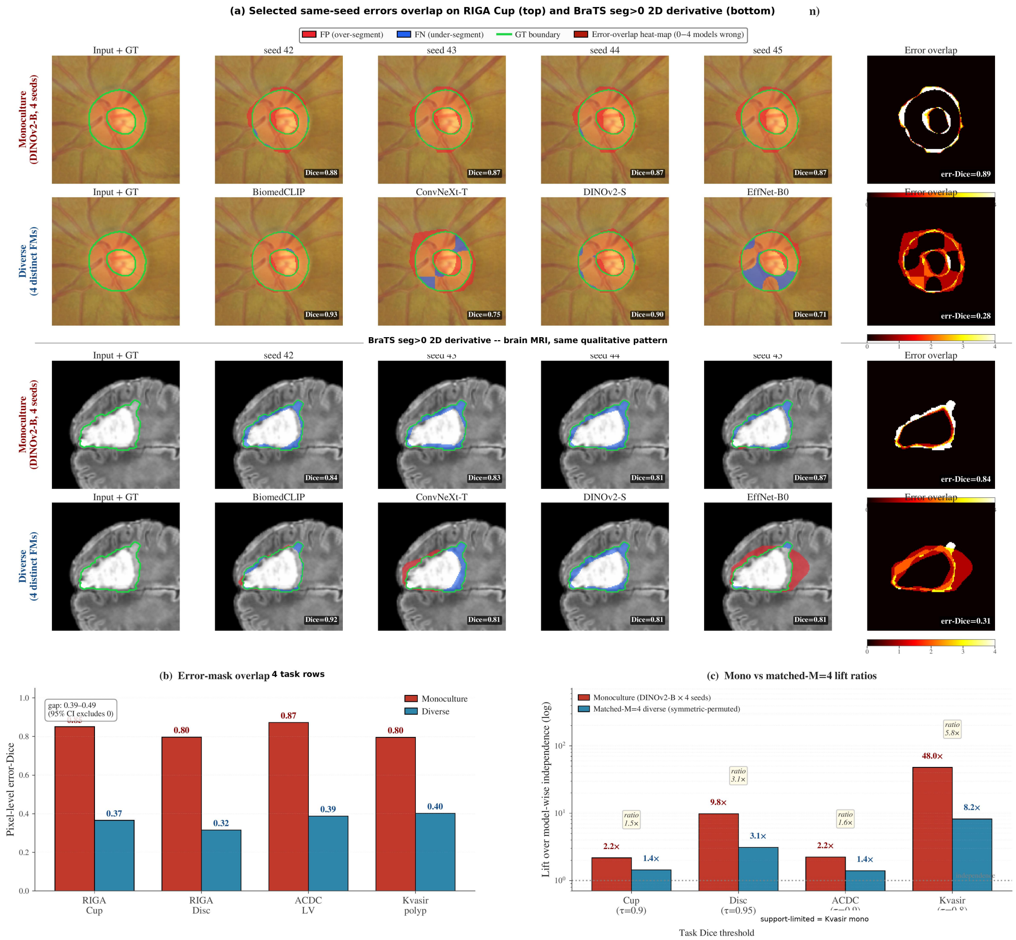

Figure 1: **Spatial and lift readouts complementing the per-case Dice gap in Table 1.** Panel (a) shows illustrative single-case visualisations on RIGA Cup and BraTS foreground (red=FP, blue=FN, green=GT boundary), comparing four DINOv2-B decoder seeds with four distinct-bundle members; cases were chosen as visual examples where same-bundle and diverse pools have comparable mean Dice but visibly different error overlap. Panel (b) reports pixel-level error-mask Dice on aggregate non-case-identical rows (RIGA Cup/Disc $N$=224, ACDC LV $N$=90, Kvasir $N$=300): same-family 0.80–0.87 vs. cross-family 0.32–0.40. Panel (c) reports selected matched-size joint-failure lift ratios. Panels (b–c) are aggregate diagnostics, not the primary per-case-Dice rows; the artifact distributes summary JSONs rather than raw masks.

Cup/Disc ($N$=224), ACDC LV ($N$=90), and Kvasir ($N$=300) rows (Fig. 1b, Appendix A3, and Appendix A8). Because these diagnostics use secondary splits or aggregate-only artifacts, we interpret them as explanatory companions to Table 1; sparse rare-event rows are descriptive rather than primary evidence.

**Item-difficulty diagnostic.** Following the warning that co-failure claims depend on the null model [Jo et al., 2026], we residualise per-case Dice against held-out item-difficulty estimates. A naive all-bundle difficulty regressor uses information from the target trace and inflates $\Delta$. A cross-fitted diagnostic estimates each trace's item-difficulty regressor from traces outside that trace's broad upstream family, then recomputes the primary same/cross rule on residuals. Because residualisation changes the correlation scale and can increase same/cross gaps, we interpret this diagnostic only qualitatively: held-out item-difficulty residualisation does not absorb the same/cross separation on any primary row (Appendix A26).

Because the primary same column is dominated by same-bundle decoder-seed pairs, we also report a same-family cross-checkpoint decomposition over the five primary rows (Appendix A30). It separates the same-bundle seed floor, distinct checkpoints within broad upstream families (DINOv2-S/B,

Table 2: **Sign-stability diagnostics for the primary rows.** Values are point estimates from distinct diagnostics and are not directly comparable in scale. Cal.-split $\Delta$ selects functional bundles on a deterministic calibration subset and evaluates the same/cross gap on disjoint cases. $P_{33}$ fail $\phi$ gap is the same/cross gap in binary failure-event correlation at the empirical 33rd percentile of model-case Dice after functional-floor filtering. Quality-control $\beta_{\mathrm{same}}$ is the same-group coefficient in a pair-level OLS diagnostic with case-bootstrap refits. The expanded-grid same-family checkpoint gap is the same-family cross-checkpoint diagnostic from Appendix A30, not the Table 1 analytic-pool cross-family stratum. Full definitions and CIs for the Table 2 columns are in Appendices A5–A7 and Appendix A30; the separate item-difficulty diagnostic is in Appendix A26.

| Task | Cal.-split $\Delta$ | $P_{33}$ fail $\phi$ gap | QC $\beta_{\mathrm{same}}$ | Checkpoint-family gap |
|---|---|---|---|---|
| Kvasir polyp | 0.250 | 0.329 | 0.169 | 0.080 |
| ACDC LV | 0.350 | 0.386 | 0.208 | 0.134 |
| BraTS foreground | 0.307 | 0.413 | 0.224 | 0.093 |
| RIGA Cup | 0.478 | 0.453 | 0.372 | 0.267 |
| RIGA Disc | 0.630 | 0.578 | 0.582 | 0.227 |

OpenAI CLIP-B/L, ResNet-18/50 when functional), and cross-family pairs. The distinct-checkpoint same-family minus cross-family gap remains positive on all five rows (0.080–0.267), while staying well below the same-bundle seed floor. Thus the primary effect is strongest for shared checkpoints and recipes, but the per-case Dice traces do not collapse to a same-decoder-seed artifact. A limited ViT same-architecture/different-pretraining diagnostic is taken up in Section 6 as a control on architecture-vs-pretraining attribution.

# 6 Same-backbone controls and mechanism limits

## 6.1 Limited ViT diagnostic: cross-pretraining correlations remain intermediate

The following diagnostics do not decompose the Table 1 gap into causal shares for architecture, pretraining corpus, checkpoint lineage, wrapper normalisation, and decoder recipe. They answer a narrower evaluation question: if a pool shares one of these upstream factors, does the trace audit still show aligned co-failure that should be reported before claiming independent failure coverage? The practical reporting recommendation does not require identifying which shared factor dominates.

A representation-similarity concern, following Kornblith et al. [2019], is that two ViTs produce similar representations *because they share architecture family*, not because they share pretraining. We hold broad architecture family fixed in a limited 5-variant ViT pool—DINOv2-B (SSL on LVD-142M), BiomedCLIP (CLIP on biomed image-text), MAE-B (ImageNet MAE), DeiT-B (ImageNet supervised), OpenAI CLIP-B—trained with the identical frozen $1{\times}1$-conv probe (Table 3). The diagnostic covers RIGA Cup and ACDC LV. In this limited diagnostic, ViT cross-pretraining correlations lie between the same-bundle seed floor and the primary cross-family reference; this ordering is descriptive only and does not identify which factor is causal.

Table 3: **Limited ViT same-architecture/different-pretraining diagnostic.** The table compares same-bundle seed-pair correlations with cross-pretraining correlations among ViT-style encoders. It is a two-task diagnostic, not a causal decomposition of architecture and pretraining. Seed floor here is the mean within-FM cross-seed Pearson restricted to the functional 5-ViT subset; this differs from Table 1's $\rho_{\mathrm{same}}$, which averages over the full 9/10-bundle post-floor pool including CNN bundles whose lower within-FM correlation pulls the mean down.

| Dataset | ViT subset | Func. ViTs | Seed floor | ViT cross-pretrain | Primary cross ref. | Seed floor − ViT cross-pretrain |
|---|---|---|---|---|---|---|
| RIGA Cup | all 5 ViTs | 5/5 | 0.968 | 0.680 | 0.361 | 0.288 |
| ACDC LV | CLIP-B dropped† | 4/5 | 0.988 | 0.807 | 0.639 | 0.182 |

Values are from the architecture-confound diagnostic and primary table summary. †The all-5 ACDC same-architecture-family ViT cross-pretraining value is 0.631 because CLIP-B is nonfunctional on ACDC (mean Dice 0.205); the functional-filtered readout is 0.807. ViT×CNN comparisons are reported only as appendix diagnostics and are not used as main-table numeric evidence.

This diagnostic alone does not isolate pretraining-bundle reuse from broad architecture-family imprint; in particular, ViT cross-pretraining correlations sit substantially closer to the same-bundle floor (gap 0.288 on RIGA Cup, 0.182 on ACDC LV) than to the primary cross-family reference,

Table 4: **Same-backbone perturbation diagnostics.** These comparisons vary decoder capacity, skip structure, or MC dropout under limited grids. They test whether simple same-backbone perturbations remove high correlation; they are not dose-matched causal controls.

| Diagnostic | Released source | Main readout | Cautious interpretation |
|---|---|---|---|
| Head-capacity sweep, RIGA Cup | Released traces, App. A31.1 | 4 heads, 769–3.94M params; Dice 0.725–0.848; cross-head same-bundle $\bar{r}$=0.690 (range 0.510–0.896) | Head design changes accuracy and some pairwise $r$, but stays above primary RIGA cross-family $\bar{r}$=0.361. |
| UNet-skip decoder replication | Released summary, App. A16 | 20 cells; Dice 0.806–0.902; seed-pair $r$=0.774–0.979 | Stronger skip head improves accuracy but does not make same-backbone pools look cross-family. |
| MC dropout, DINOv2-B/RIGA Cup | Released traces, App. A31.7 | MC $\bar{r}$=0.947, 0.925, 0.893 for $p$=0.1, 0.2, 0.4 | Weak decorrelation lever; largest tested drop is about 0.10. |

consistent with shared architecture and shared $1 \times 1$-conv readout contributing materially to the measured dependence.

The released CNN random-init diagnostic (Appendix A31.2) shows positive same-vs-cross structure even without shared pretrained weights ($\Delta = 0.086$ on ACDC LV, $\Delta = 0.121$ on RIGA Cup); a more substantive 3-architecture from-scratch replication (Appendix A13) reaches the same conclusion. The pretrained-bundle effect therefore operates on top of, not in isolation from, an architecture-and-recipe baseline.

### 6.2 Decoder-capacity, stochastic, and medical-domain probes

**(C1) Head architecture/capacity.** The head-capacity and UNet-skip decoder perturbation diagnostics change accuracy and reduce some pairwise correlations, but their same-backbone correlations remain above the primary RIGA cross-family reference. Thus the tested decoder-capacity changes did not remove same-backbone co-failure in these diagnostics.

**(C2) Stochastic/adaptation levers.** In the RIGA Cup diagnostic, MC averaging modestly lowers within-bundle correlation but remains far above the cross-family reference. Bootstrap-bagging and fold-reslicing are appendix diagnostics with different sampling surfaces; the fold-reslicing row is a single-seed split-stability check, not a within-fold versus cross-fold same-backbone estimator. Full-fine-tuning and LoRA probes without trace artifacts are described only to delimit scope and are not used to support the main claims.

**(C3) Medical-domain encoder-only probe.** A MedSAM image-encoder-only sensitivity is described in Appendix A10 as a summary diagnostic rather than primary evidence. The probe strips MedSAM's prompt encoder and mask decoder, so it does not evaluate prompt-conditioned MedSAM. RETFound is documented only as a gated-checkpoint path; without access to the gated checkpoint, we do not report RETFound rows or use them to support the claims.

## 7 Evidence taxonomy and scope boundaries

Same-backbone perturbations are useful but incomplete: bootstrap-bagging reduces within-pool $\bar{r}$ by only 0.05–0.12, fold-reslicing keeps cross-bundle $\bar{r}$ stable but has one seed per fold, and MC dropout remains a high-correlation same-bundle readout (Appendices A14, A17, A31.7). The same-architecture-family ViT and CNN random-init readouts (Appendices A31.2, A29, A13) indicate that pretraining, architecture, and shared recipe effects remain entangled rather than collapsing to a single mechanism. The main scope boundary is full-pipeline segmentation: nnU-Net v2 2D/3D complements give task-level different-fold gaps of $\Delta \approx 0.034$–$0.047$ on RIGA Cup and $\Delta \approx 0.043$ on ACDC 3D, and a case-identical RIGA Cup nnU-Net check gives $\Delta$=0.008 [0.002, 0.039] with both correlations very high (0.984 vs. 0.976); these are descriptive scope checks, not replacements for the primary audit. A consolidated lever map is in Appendix A21, and Appendix A24 reports a 3D segmentation-architecture pilot. Appendix Table 38 encodes the tier structure: Table 1 is the primary frozen-encoder audit, while estimator, event, architecture/pretraining, same-backbone, full-pipeline, and held-out rows are diagnostic or scope evidence.

## 8 Discussion

**Practical reading.** Report $M_{\text{eff}}$ as a threshold-specific dependence readout before treating same-bundle seeds as independent evidence, not as a standalone pool-selection objective. In our released rows, diverse $M{=}4$ pools yield mean $M_{\text{eff}}$ from 2.05 (Kvasir) to 2.97 (RIGA Disc) under the paper's equicorrelation readout at $\tau = 0.85$, rather than 4; using the maximum observed diverse-pool $\rho_{\text{fail}}$ gives conservative lower readouts of 1.30–1.69 for the most redundant valid 4-bundle tuples. Same-bundle seed pools yield $M_{\text{eff}}$ near 1 under the same threshold. The readout should always be paired with marginal Dice and matched-quality comparisons before any pool-selection decision (Appendix A27). For procurement-style decisions where a clinical-imaging team is choosing between an $N$-bundle FM pool and a smaller curated pool, an in-domain co-failure audit on a small validation set should precede pool-size-based diversity claims.

**Implications for ensemble methodology.** Standard nnU-Net 5-fold validation under shared encoders is a within-bundle estimator and inherits the dependence structure measured here; cross-encoder ensembles should be preferred where co-failure is decision-critical. A structured co-failure audit reporting $\Delta$, joint-failure lift, and $M_{\text{eff}}$ together provides one such readout, and Appendix A28 gives a random-effects representation that motivates why same-backbone fold/seed surfaces concentrate above cross-encoder references.

**Limitations.** The evidence is retrospective and benchmark-based: this study audits a reporting assumption about failure evidence, not clinical deployment, and we do not claim population-level inference over future encoders, datasets, or recipes. The BraTS foreground row is a 2D derivative of the 2024 GLI release, not the official 3D challenge target. Table 1 is the confirmatory evidence surface; threshold, lift, and held-out rows are exploratory diagnostics. Per-case Dice Pearson is a continuous-monotone alignment statistic, so two members can have high Pearson Dice yet still disagree on which low-quality cases fail; pixel error-mask Dice and joint-failure lift in Section 5 supplement it at the failure-event level, and Geirhos-style binarised error consistency would be a complementary discrete estimand. The cross-bundle stratum draws from at most 8 broad family clusters across 11 bundles, so few-cluster bias is unavoidable; we report case-level and hierarchical bootstrap CIs jointly. The frozen $1{\times}1$-conv decoder is deliberately minimal so the encoder dominates the readout; absolute correlation magnitudes are upper bounds for richer decoders, and the M3 head-capacity and M4 UNet-skip diagnostics show more capable heads reduce some pairwise correlations but do not eliminate the same-vs-cross gap. All ten sign tests (five rows $\times$ two CI methods) independently exclude zero, with the smallest hierarchical lower bound at 0.174 (Kvasir); we report directional gaps per row rather than a joint significance statement. Appendices A18, A19, and A15 extend these caveats.

**What the evaluation supports.** The E&D takeaway is a reporting requirement: when a medical segmentation pool reuses pretrained encoder checkpoints, input statistics, and decoder recipes, aggregate Dice should be accompanied by aligned case-level co-failure correlations before the pool is credited with independent failure coverage. The result is a finite audit of evaluation evidence, not a clinical safety claim or a causal decomposition of pretraining and architecture.

**Release.** The anonymized artifact is available through the supplementary submission and anonymous code URL `https://anonymous.4open.science/r/NIPS_A_ACF-86CD`. It includes primary per-case Dice JSONs, summary tables, code, metadata/manifests, hashes, environment files, a data card for the derived trace resource, selected appendix summary JSONs, and Croissant metadata including the required Responsible AI metadata fields for the derived trace/code artifact. It does not introduce a new raw dataset; the released JSONs are derived trace artifacts that retain upstream de-identified alignment identifiers and source labels, including subject/patient/slice IDs and deterministic hashed RIGA alignment IDs, solely for deterministic alignment and subject-level resampling. Released IDs are not intended for individual-level analysis, subgroup profiling, re-identification, inference of protected attributes, clinical diagnoses, or patient outcomes. The bundle does not include raw medical images, raw masks, direct identifiers, full prediction caches, or held-out raw prediction arrays; raw data and checkpoints remain governed by upstream licenses and DUAs.

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# A1 Appendix roadmap and evidence inventory

The appendix is organised by evidentiary role rather than by experiment chronology. It starts with diagnostic readouts, then documents primary estimators, uncertainty, task-level sources, and estimator sensitivity. Middle sections collect same-backbone perturbations, segmentation-pipeline scope checks, distribution-shift and held-out stress tests, operational metrics, and $M_{\text{eff}}$. Later sections collect mechanism diagnostics and additional scope tables, and the appendix ends with the release manifest and artifact metadata. Secondary rows are retained only when they qualify or explain a paper claim, and their captions state when they use non-primary pools or splits. Unless explicitly marked primary or sourced from the primary `results/_merged` summaries, appendix values are diagnostic rather than primary support for Table 1.

**CLIP-L notation.** Throughout the appendix, CLIP-L/14 refers to the preserved OpenCLIP `ViT-L-14/openai` dense-token wrapper used to generate the released traces. It is not a claim of OpenAI `clip.load` parity; a strict QuickGELU-parity CLIP-L loader would be a separate sensitivity with a distinct FM identifier.

Table 5: **Appendix guide.** Blocks describe the actual order of sections below; cross-references point to the detailed tables rather than repeating them in the main paper.

| Block | Sections (in order) | Evidence role |
|---|---|---|
| Estimators and provenance | App. A2, A3, A4, A5, A6 | Diagnostic readouts, pixel overlap, case/hierarchical bootstrap, functional-floor/ICC/permutation sensitivity, pair-quality OLS. |
| Operational metrics | App. A7, A8, A9, A10 | Failure-event $\tau$-sweep, conditional recovery, Kvasir secondary split, matched-$M$ lift, MedSAM probe details. |
| Same-backbone controls | App. A11, A12, A13, A14 | Subject clustering, BraTS sparse caveat, ACDC architecture caveat / random-init / LoRA scope, bootstrap-bagging. |
| Adaptation & distribution shift | App. A15, A16, A17, A18 | RIGA cross-vendor, UNet-skip decoder, nnU-Net fold reslicing + case-identical, segmentation-recipe scope (loss/aug, length, dimensionality). |
| Held-out and meta-analyses | App. A19, A21, A22, A23, A24, A25 | Held-out 5-task stress test, lever index, leave-one-family-out, lift-ceiling support, 3D pilot, matched-recipe adaptation scope. |
| Mechanism diagnostics | App. A26, A27, A28, A29, A30 | Item-difficulty residualisation (cross-fitted + holdout), $M_{\text{eff}}$, random-effects representation, ViT cross-pretraining, cross-checkpoint same-family. |
| Additional scope diagnostics | App. A31, A31.1, A31.2, A31.3, A31.4, A31.5, A31.6, A31.7 | Decoder capacity, random-init CNN, loss/aug sensitivity, fold CV, BraTS train-size, BraTS multimodal, MC dropout. |
| Artifact record | App. A32 | Bundle manifest, claim-to-artifact map, compute scope, trace inventory. |

# A2 Diagnostic readouts and scope

This section records diagnostic readouts that clarify scope without adding new primary claims.

**RIGA annotation source.** Table 1 uses the six-rater majority consensus as ground truth for both RIGA Cup and RIGA Disc, matching the released RIGA per-case Dice traces. RIGA also distributes individual ophthalmologist masks, but individual-rater replay artifacts are not included in the supplementary artifact. The consensus Cup and Disc rows are therefore the only RIGA annotation-source evidence used by the manuscript.

**Probability-map diagnostics.** The artifact releases per-case Dice traces and selected aggregate pixel/lift diagnostics, but not raw probability maps. We therefore do not use mean-confidence, probability-calibration, or failure-flagging AUROC values as evidence for any paper claim.

## A3 Pixel-level error-overlap breakdown

Table 6: **Pixel-level error-overlap diagnostic on selected secondary rows.** Values are same/cross gaps in error-mask overlap on aggregate diagnostic splits, not primary case-level evidence. For error-Dice, no bootstrap replicate out of $B{=}500$ produced a non-positive gap on any task cell, giving the finite-$B$ bound $\Pr[\text{gap}{\leq}0] < 1/501$; FP/FN/boundary rows are point-estimate breakdowns.

| Metric | RIGA Cup | RIGA Disc | ACDC LV |
|---|---|---|---|
| Error-Dice gap | 0.485 | 0.481 | 0.485 |
| Error-Jaccard gap | 0.515 | 0.481 | 0.521 |
| FP-only error-Dice gap | 0.453 | 0.494 | 0.566 |
| FN-only error-Dice gap | 0.527 | 0.464 | 0.449 |
| Boundary-band (5px) | 0.430 | 0.439 | 0.433 |

FP = (pred foreground, GT background); FN = (pred background, GT foreground); boundary-band restricts error masks to pixels within 5 pixels of the GT edge (trimap). All metrics are pair-wise Dice between the two models' error masks (or empty-mask-excluded Jaccard). On RIGA Cup, RIGA Disc, and ACDC LV binary, the similar-magnitude gaps across FP, FN, and boundary variants indicate the spatial co-failure is not driven by a single error type on those three tasks (Kvasir and BraTS not tabulated in this split).

## A4 Bootstrap sensitivity of the gap

Table 1 reports 95% CIs from a three-level nested hierarchical bootstrap that resamples family labels, seeds within each family, and cases ($B{=}2000$). We additionally compute a *case-only* bootstrap that fixes the family/seed structure and resamples the $N$ test cases with replacement ($B{=}2000$). The two procedures target different sampling dimensions: the hierarchical CI jointly perturbs the finite FM/seed/case surface, whereas the case-only CI asks "what if we had a different test set" conditional on the evaluated pool. We report both as sensitivity evidence for the same finite audit rather than as population-level guarantees.

Case-only bootstrap intervals match the Case CI column of Table 1 exactly (the case-only resampling uses the same fixed family/seed structure as the per-row case CI), so we do not retabulate them here. The remainder of this section reports BraTS pool reconciliation and the slice-vs-subject sampling caveats.

**BraTS foreground pool reconciliation.** The 5-FM, 8-FM, and 8-FM-alt-estimator BraTS values cited elsewhere differ because they use disjoint analytic pools (5-FM: a subset of the 11-bundle source pool; 8-FM: a completed-only subset; alt-estimator: scoring with a different family-grouping convention). The principled headline is the post-floor 10-FM value because it uses the Dice$\geq$0.30 functional floor and the Table 1 same/cross definition consistent with every other primary row. The primary BraTS foreground result is the 10-bundle Table 1 value of $\Delta{=}0.315$ [0.302, 0.329] (case-level CI), from the 11-bundle source pool with 10 retained bundles after functional-floor filtering (BiomedCLIP, CLIP-B/16, CLIP-L/14, ConvNeXt-T, DeiT-B/16, DINOv2-B/S, EfficientNet-B0, MAE-B/16, ResNet-50; ResNet-18 is below the Dice floor). Source: `results/_merged/paper_table.json`. Subset diagnostics without released aggregate files are omitted from the released numeric evidence. The value reported throughout the main text and abstract is therefore the Table 1 value.

All gap CIs strictly exclude zero under the case-level bootstrap, matching the family/seed-level CIs in the main text.

**Per-rater RIGA Cup sensitivity (legacy non-replayable pool).** A non-replayable legacy sensitivity pool (containing RETFound traces not in the released artifact) repeated the symmetric multi-seed analysis using each of RIGA's six individual rater masks; per-rater Pearson within–cross gaps were $0.46$–$0.53$ across all six raters, with the range across raters an order of magnitude smaller than the within-vs-cross gap itself. The released Table 1 RIGA Cup row remains the bundle-replayable evidence.

## A5 Estimator-robustness and negative-control checks

The primary row uses Pearson correlation and a Dice$\geq 0.30$ per-FM functional floor. To make this choice auditable, the released bundle includes a sensitivity artifact, `results/_merged/diagnostics/audit_sensitivity.json`, generated from the same per-case Dice JSONs by `code/scripts/compute_audit_sensitivity.py`. It recomputes the within–cross gap under a floor sweep, an ICC(A,1) alternative, leave-one-FM-out summaries, leave-one-task-out meta summaries, and a random family-label permutation control.

Table 7: **Estimator sensitivity on primary traces.** Pearson values are the Table 1 point estimates at the Dice$\geq 0.30$ floor. ICC(A,1) is an alternate agreement statistic on the same retained members. The floor range reports the minimum/maximum Pearson gap over $\{0, 0.20, 0.25, 0.30, 0.35, 0.40\}$ where the pool is non-empty. The permutation note reports a non-exchangeable random family-label stress test over completed members, using $(1 + \#\{\Delta_{\text{null}} \geq \Delta_{\text{obs}}\})/(1001)$; this is not interpreted as a $p$-value. Retained counts at every floor are in `audit_sensitivity.json`.

| Task | Members at 0.30 | Pearson gap | ICC(A,1) gap | Floor-sweep gap range |
|---|---|---|---|---|
| Kvasir | 44 | 0.261 | 0.333 | $0.261^\flat$ |
| ACDC LV | 40 | 0.321 | 0.401 | 0.242–0.355 |
| BraTS foreground | 40 | 0.315 | 0.404 | 0.315–0.362 |
| RIGA Cup | 36 | 0.473 | 0.539 | 0.384–0.529 |
| RIGA Disc | 44 | 0.638 | 0.716 | $0.638^\flat$ |

All five rows have observed-gap-exceeded rank fraction $0.001$ at the 0.30 floor. $^\flat$ Value is constant across all valid floors in $\{0, 0.20, 0.25, 0.30, 0.35, 0.40\}$.

The five-task mean Pearson gap at the paper floor is $0.401$ (median $0.321$, range $0.261$–$0.638$). Leave-one-task-out means stay in $0.342$–$0.437$; the smallest value occurs when the largest-gap RIGA Disc row is held out. These checks do not add new cases or new model training. They instead reduce the risk that the primary sign is an artifact of the exact functional floor, the Pearson statistic, or arbitrary family labels.

**Calibration-selected functional pools.** The primary table defines a functional pool using all released evaluation traces. To check whether the sign depends on this convention, the scorer also makes a deterministic split of each task's aligned units, selects bundles by mean Dice on the calibration subset, and evaluates the same/cross gap on disjoint cases. This is a leakage-sensitivity analysis rather than a prospective validation protocol, but it addresses the specific concern that the functional-floor sign is induced by evaluating the floor and the gap on exactly the same cases.

Table 8: **Calibration-selected functional-pool sensitivity.** Functional bundles are selected on the calibration subset and evaluated on disjoint cases under the Table 1 same/cross grouping. `Unit-agg.` $\Delta$ is reported only for nested slice datasets after patient/subject aggregation. ACDC can select 11 bundles here because selection is calibration-only, whereas Table 1 applies the retrospective all-evaluation functional floor. CIs are case-bootstrap intervals on the evaluation subset. Source: `results/_merged/calibration_split/*.json`, regenerated by `code/scripts/compute_all.py`.

| Task | Cal./eval units | Selected bundles | Eval. $\Delta$ [95% CI] | Unit-agg. $\Delta$ |
|---|---|---|---|---|
| Kvasir | 42/78 images | 11 | 0.250 [0.196, 0.322] | – |
| ACDC LV | 7/13 patients | 11 | 0.350 [0.313, 0.392] | 0.206 |
| BraTS foreground | 113/211 subjects | 10 | 0.307 [0.292, 0.323] | 0.205 |
| RIGA Cup | 33/62 images | 9 | 0.478 [0.369, 0.599] | – |
| RIGA Disc | 33/62 images | 11 | 0.630 [0.534, 0.695] | – |

## A6 Quality-controlled pairwise-correlation diagnostic

A possible alternative explanation is that same-FM pairs have higher correlation merely because they have matched marginal quality. We therefore add a CPU-only diagnostic that uses the same released per-case Dice traces retained by Table 1 and fits a pair-level OLS model:

$$r_{ab} \sim I_{\text{same}} + \bar{D}_{ab} + |\Delta \bar{D}_{ab}| + \bar{V}_{ab} + |\Delta V_{ab}|.$$

Here $\bar{D}_{ab}$ and $\bar{V}_{ab}$ are the pair mean Dice and average per-case Dice variance, with absolute pair differences as additional controls. The same-group indicator is exactly the primary grouping rule: same FM plus the DINOv2-B/S pair. The coefficient is not a causal estimand, but it checks that the main sign is not removed by simple pair-quality and variance controls. The diagnostic is generated by `code/scripts/compute_quality_controlled_pairs.py` and released as `results/_merged/diagnostics/quality_controlled_pairs.json`.

Table 9: **Pair-quality controlled diagnostic.** $\beta_{\text{same}}$ is the coefficient on the primary same-group indicator in a pair-level OLS check. Same-group coefficients remain positive after controlling for pair mean Dice, absolute mean-Dice difference, average per-case variance, and absolute variance difference. CIs are case-bootstrap intervals from $B{=}1000$ refits and should be read as diagnostic, not causal, uncertainty.

| Task | Members | Same pairs | Cross pairs | $\beta_{\text{same}}$ [95% CI] |
|---|---|---|---|---|
| Kvasir | 44 | 82 | 864 | 0.169 [0.124, 0.220] |
| ACDC LV | 40 | 76 | 704 | 0.208 [0.175, 0.242] |
| BraTS foreground | 40 | 76 | 704 | 0.224 [0.210, 0.239] |
| RIGA Cup | 36 | 70 | 560 | 0.372 [0.297, 0.456] |
| RIGA Disc | 44 | 82 | 864 | 0.582 [0.495, 0.639] |

## A7 Failure threshold sensitivity

The primary estimand is continuous per-case Dice correlation. To make the co-failure interpretation less dependent on continuous-score correlation, the released artifact also recomputes the Table 1 same/cross grouping on binary failure events. Table 10 reports the empirical $P_{33}$ readout across all five primary task rows; the released JSON also contains $P_{25}$ and $P_{50}$ thresholds. The RIGA Cup all-fail table below is retained as a pool-level threshold diagnostic, and absolute-threshold lift with support flags is in Appendix A9.

Table 10: **Binary failure-event diagnostic at the empirical $P_{33}$ threshold.** Failure is defined as per-case Dice below the 33rd percentile of all model-case Dice values after functional-floor filtering. $\phi$ is binary-event Pearson correlation averaged over same-group or primary cross-group pairs. Joint fail same/cross is the pairwise joint-failure probability averaged over the corresponding model-trace pairs. CIs are case-bootstrap intervals for the same–cross $\phi$ gap. This is not a clinically chosen threshold. Source: `results/_merged/diagnostics/failure_event_summary.json`, regenerated by `code/scripts/compute_failure_event_summary.py`.

| Task | $\tau_{P33}$ | Same $\phi$ | Cross $\phi$ | Gap [95% CI] | Joint fail same/cross |
|---|---|---|---|---|---|
| Kvasir | 0.659 | 0.869 | 0.540 | 0.329 [0.278, 0.388] | 27.9/21.6% |
| ACDC LV | 0.548 | 0.875 | 0.489 | 0.386 [0.354, 0.418] | 27.7/19.9% |
| BraTS foreground | 0.636 | 0.838 | 0.425 | 0.413 [0.401, 0.425] | 27.3/18.7% |
| RIGA Cup | 0.575 | 0.706 | 0.253 | 0.453 [0.399, 0.509] | 24.4/14.4% |
| RIGA Disc | 0.857 | 0.703 | 0.125 | 0.578 [0.511, 0.631] | 24.8/12.0% |

**Full $\tau$-sweep of conditional recovery.** Table 11 tabulates grand-mean conditional recovery rates $P[\text{pool-mean Dice} \geq \tau \mid \text{any member Dice} < \tau]$ for same-bundle pools (4 seeds of one FM, averaged over all FMs passing the per-task Dice $\geq$ 0.30 functional floor) and diverse pools (all $\binom{K}{4}$ functional-subset compositions and all released seed tuples) at $\tau \in \{0.70, 0.75, 0.80, 0.85, 0.90\}$. Source: `results/_merged/diagnostics/tau_sweep.json`, regenerated by `code/scripts/compute_tau_sweep.py`. Recovery is $\tau$-fragile in both regimes; the diverse $-$ same-bundle gap (column $\Delta$) is non-monotone in $\tau$, so the table motivates auditing threshold choices rather than claiming monotonic operational superiority of diverse pools.

Table 11: **Secondary $\tau$-sweep of conditional recovery rate $P$[ens pass | any fail].** Grand-mean across all functional 4-of-$K$ diverse compositions and all functional same-bundle baselines. Recovery gaps vary with threshold and can become near-zero or negative near ceiling; these values are operational diagnostics, not primary support for the same/cross correlation claim.

| Task | $\tau$ | Mono | Diverse | $\Delta$ |
|---|---|---|---|---|
| RIGA Cup (9-bundle) | 0.70 | 0.061 | 0.214 | +0.153 |
| | 0.75 | 0.073 | 0.120 | +0.047 |
| | 0.80 | 0.055 | 0.050 | −0.005 |
| | 0.85 | 0.010 | 0.007 | −0.003 |
| | 0.90 | 0.001 | 0.000 | −0.001 |
| ACDC LV (10-bundle) | 0.70 | 0.074 | 0.285 | +0.211 |
| | 0.75 | 0.062 | 0.237 | +0.175 |
| | 0.80 | 0.049 | 0.163 | +0.114 |
| | 0.85 | 0.040 | 0.072 | +0.032 |
| | 0.90 | 0.020 | 0.012 | −0.007 |
| Kvasir-SEG (11-bundle) | 0.70 | 0.092 | 0.388 | +0.295 |
| | 0.75 | 0.074 | 0.360 | +0.285 |
| | 0.80 | 0.058 | 0.267 | +0.209 |
| | 0.85 | 0.050 | 0.156 | +0.106 |
| | 0.90 | 0.025 | 0.030 | +0.005 |
| BraTS foreground (2D FLAIR `seg>0`) | 0.70 | 0.080 | 0.326 | +0.245 |
| | 0.75 | 0.061 | 0.233 | +0.172 |
| | 0.80 | 0.039 | 0.129 | +0.090 |
| | 0.85 | 0.019 | 0.035 | +0.016 |
| | 0.90 | 0.004 | 0.001 | −0.003 |

At the strictest tested threshold ($\tau$=0.90), the diverse advantage in conditional recovery vanishes and reverses on RIGA Cup ($\Delta = -0.001$) and ACDC LV ($\Delta = -0.007$), while remaining positive on Kvasir ($+0.005$) and slightly negative on BraTS ($-0.003$). The diverse-pool benefit is concentrated in the moderate-failure regime ($\tau \in [0.70, 0.80]$), consistent with the primary correlation gap predicting reduced *but not zero* coverage benefit at extreme thresholds.

Table 12: **RIGA Cup threshold diagnostic.** Values are for the released primary RIGA Cup functional pool only. Mono pools are same-bundle four-seed pools averaged over functional FMs; diverse pools are four distinct functional bundles. Percentile thresholds are computed over all model-case Dice values after the Dice≥0.30 functional floor. Source: `results/_merged/diagnostics/threshold_table_riga_cup.json`.

| Threshold | Mono fail $\bar{r}$ | Diverse fail $\bar{r}$ | Mono ALL-fail | Diverse ALL-fail |
|---|---|---|---|---|
| $P_{25}$ ($\tau$=0.513) | 0.833 | 0.209 | 19.9% | 2.2% |
| $P_{33}$ ($\tau$=0.575) | 0.803 | 0.257 | 26.5% | 5.3% |
| $P_{50}$ ($\tau$=0.686) | 0.840 | 0.268 | 44.8% | 15.4% |

## A8 Kvasir secondary split diagnostic

Pixel-level diagnostic on this split: mono 0.80 vs. diverse 0.40, gap 0.394 [95% CI 0.38, 0.40]. Absolute-threshold lift at Dice<0.7: mono 252× vs. diverse 120× (ratio 2.1×); at Dice<0.8: 48.4× vs. 14.6× (3.3×); at Dice<0.9: 4.6× vs. 1.8× (2.5×). The numbers are slightly smaller than RIGA/ACDC, likely reflecting the larger test set (300 cases vs. 224 / 90) and more variable polyp presentations, but qualitatively identical: mono still shows substantially higher correlated-failure lift than diverse.

## A9 Matched-$M$ lift comparison

To ensure the mono vs. diverse comparison is not driven by pool size, we construct an *oracle* matched-$M$=4 diverse pool by selecting the top-4 bundles by *test-set seed-averaged* mean Dice (rank computed across all four decoder seeds, avoiding seed-42 selection circularity; selected bundles per task are recorded in the appendix diagnostic output—e.g. Kvasir top-4 = {DINOv2-

Table 13: **Kvasir-SEG secondary split diagnostic.** This generic 8-bundle summary uses a 1,000-image, 700/300 train/test split and is not the 11-bundle, 880/120 Kvasir row in Table 1. Seven of eight generic encoder bundles produce functional segmentation (Dice$>0.40$); CLIP ViT-L/14 is non-functional and excluded from cross-family analysis. MedSAM-only traces are reported separately as an encoder-probe summary in Appendix A10; no RETFound Kvasir trace is included in the released artifact bundle.

| FM Family | Dice | Within $\bar{r}$ | Status |
|---|---|---|---|
| DINOv2 ViT-B/14 | 0.819 | 0.972 | functional |
| DINOv2 ViT-S/14 | 0.807 | 0.951 | functional |
| BiomedCLIP ViT-B/16 | 0.717 | 0.954 | functional in this split |
| ConvNeXt-Tiny | 0.651 | 0.949 | functional |
| EfficientNet-B0 | 0.582 | 0.945 | functional |
| ResNet-50 | 0.540 | 0.880 | functional |
| ResNet-18 | 0.468 | 0.876 | functional |
| CLIP ViT-L/14 | 0.174 | 0.839 | non-functional |

B, DINOv2-S, BiomedCLIP, ConvNeXt-T}, Cup top-4 = {BiomedCLIP, DINOv2-B, DINOv2-S, EfficientNet-B0}). This is deliberately favourable to the diverse pool and useful for a matched-size structural comparison, but it is not a deployable pool-selection rule because the ranking uses the same test set on which lift is evaluated. For each task / threshold we report three lift variants:

- **Matched-4 seed-42 (single-seed diagnostic)**: the single-seed pool; retained for comparability with a single-seed estimator.
- **Matched-4 symmetric permuted (preferred)**: each of the 4 FMs takes a *distinct* decoder seed from $\{42, 43, 44, 45\}$; the lift is averaged on the log scale over all $4! = 24$ seed-to-FM permutations, then exponentiated. This is the matched-size structural analog of the main paper's symmetric Pearson $\bar{r}$ estimator (mono = DINOv2-B with 4 distinct seeds matches matched-4 = 4 FMs with 4 distinct seeds).
- **Matched-4 symmetric independent (sensitivity)**: each FM's seed is drawn i.i.d. with replacement from $\{42..45\}$, $4^4 = 256$ tuples; reported as a robustness check (the permuted variant is preferred because mono uses four *distinct* seeds).

All bootstrap CIs are computed on *cases* only (patient-clustered on ACDC); the seed tuple space is finite and fully enumerated, not bootstrapped. We report 95% case-bootstrap CIs on the log-scale lift and exponentiate to the reported interval ($B$=2000). We also report the raw all-fail count ($\mathrm{AF_{cnt}}$) and the Poisson-binomial independence expectation $\mathbb{E}[\mathrm{AF_{ind}}]$; when $N \cdot \mathbb{E}[\mathrm{AF_{ind}}] < 5$ we flag the row as support-limited. Under the matched-size comparison, the adequately supported rows are RIGA Cup $\tau$=0.9, RIGA Disc $\tau$=0.95, and ACDC LV $\tau$=0.9; Kvasir $\tau$=0.8 is mono-sparse and retained as a descriptive diagnostic to match Fig. 1. Across these selected rows the mono/matched-4 lift ratio is 1.5–5.8$\times$ (Table 14); lower-support rows give larger ratios (5.9–17$\times$) treated as descriptive order-of-magnitude only. We therefore treat $\tau$ values well below the task-specific Dice ceiling as support-limited and report them as descriptive rather than precise lift estimates.

Table 14: **Matched-$M$=4 lift diagnostic.** Mono is DINOv2-B $\times$ 4 seeds; matched-4 diverse uses the top-4 bundles by test-set seed-averaged mean Dice, so this is an oracle structural comparison, not a deployable selection rule. CIs are 95% case-bootstrap intervals on log-lift, exponentiated. Rows without adequate support (✓ missing) are descriptive order-of-magnitude indicators.

| Task | $\tau$ | Mono [CI] | Diverse-42 | Diverse-perm [CI] | Ratio* | Supp. |
|---|---|---|---|---|---|---|
| RIGA Cup | 0.8 | 4450 [1408, 24012] | 174 [79, 467] | **265 [130, 696]** | 17× | – |
| RIGA Cup | 0.9 | 2.18 [1.81, 2.69] | 1.33 [1.20, 1.47] | **1.44 [1.31, 1.60]** | 1.5× | ✓ |
| RIGA Disc | 0.95 | 9.80 [6.82, 14.6] | 2.87 [2.19, 3.82] | **3.11 [2.44, 3.98]** | 3.2× | ✓ |
| ACDC LV | 0.5 | 2149 [462, 34449] | 200 [51, 1026] | **188 [62, 817]** | 11× | – |
| ACDC LV | 0.7 | 189 [55.6, 1064] | 31.2 [15.5, 77.7] | **25.3 [13.1, 57.8]** | 7.5× | – |
| ACDC LV | 0.8 | 33.8 [15.1, 107] | 5.87 [3.60, 10.6] | **5.71 [3.63, 9.93]** | 5.9× | – |
| ACDC LV | 0.9 | 2.23 [1.63, 3.32] | 1.39 [1.22, 1.60] | **1.40 [1.25, 1.61]** | 1.6× | ✓ |
| Kvasir | 0.7 | 246 [128, 570] | 33.9 [22.3, 54.5] | **30.8 [20.4, 48.3]** | 8.0× | – |
| Kvasir | 0.8 | 48.0 [28.9, 85.1] | 8.72 [6.62, 12.0] | **8.21 [6.35, 11.1]** | 5.8× | mono-sparse |

* Ratio = Mono / Matched-4 PERM. **Bold** = preferred symmetric-permuted estimator (24 seed-to-FM tuples). Matched-4 seed 42 is a single-seed diagnostic. Matched-4 INDEP ($4^4$=256 iid tuples) produces values within $\pm 1.5\%$ of PERM on every row and is omitted for compactness; the full table is a secondary diagnostic rather than the `results/_merged` primary scorer output. Supp. ✓ iff $N \cdot \mathbb{E}[\mathrm{AF_{ind}}] \geq 5$ for the PERM estimator. Mono lift values span four orders of magnitude (1.4–4,450$\times$); precision is reported to the data's underlying decimals and is not uniform across the lift column.

## A10 Medical-domain encoder probe details

We implemented two medical-domain encoder paths, but only the MedSAM image-encoder probe is reported in the released evidence bundle:

**RETFound** is a ViT-L/16 MAE pretrained on 1.6M retinal images [Zhou et al., 2023]. The code path builds the RETFound-MAE CFP ViT-L/16 trunk and expects the gated HuggingFace checkpoint `YukunZhou/RETFound_mae_natureCFP` or a user-specified checkpoint path; the official upstream project is `rmaphoh/RETFound_MAE`. RETFound MAE pretraining uses ImageNet $[0.485, 0.456, 0.406]/[0.229, 0.224, 0.225]$ normalisation; our frozen-probe adaptation would drop the CLS token and reshape the $196{=}14{\times}14$ patch tokens into a spatial $14{\times}14$ feature map of dimension 1024. That patch-token extraction is not RETFound's official global/CLS-style classification fine-tuning path. Because the gated checkpoint was unavailable in the release environment, RETFound cells are not reported and no RETFound result supports the paper's empirical claims.

**MedSAM** is a SAM ViT-B fine-tuned on 1.5M medical image-mask pairs [Ma et al., 2024]. The code path uses the official `bowang-lab/MedSAM` `medsam_vit_b.pth` checkpoint through `segment_anything.sam_model_registry['vit_b']` with `MEDSAM_CHECKPOINT` pointing to that checkpoint. In this audit, we use *only the image encoder* as a frozen feature extractor with our same $1{\times}1$-conv decoder. We discard MedSAM's prompt encoder and mask decoder entirely—i.e. MedSAM enters this boundary probe on the same frozen-feature terms as the other FMs (no prompt-based path, no bounding-box input, no iterative refinement). The wrapper strips the SAM image-encoder neck and uses the pre-neck ViT-B representation: the official $1024{\times}1024$ positional grid is bicubically interpolated to the $224{\times}224$ input grid, yielding a $14{\times}14$ spatial feature map of dimension 768. This pre-neck $768{\times}14{\times}14$ tensor is an internal representation, not the official SAM/MedSAM post-neck image embedding at 1024 input. This means MedSAM in our experiment is a "MedSAM image-encoder probe", not its intended prompt-conditioned segmentation system or official inference preprocessing path. The comparison is therefore an encoder-only boundary probe for the same-bundle dependence question we ask, and it does not evaluate MedSAM as released or its best-case performance with its own prompt-based decoder.

All other aspects of the secondary sensitivity use the same four decoder seeds, 30-epoch Adam $10^{-3}$ training, and identical $1{\times}1$ conv decoder. The completed MedSAM-only probe covers ACDC LV, Kvasir, RIGA Cup, and RIGA Disc (four seeds each). Mean Dice across seeds is 0.517, 0.513, 0.734, and 0.908 respectively; the aggregate diagnostic is released as `results/_merged/diagnostics/medsam_only_summary.json`. We use this as scope context rather than primary evidence.

## A11 BraTS/ACDC clustering sensitivity

The Table 1 BraTS foreground row is a BraTS 2024 GLI 2D FLAIR derivative with $n{=}3{,}228$ test slices from 324 held-out subjects: per held-out subject, the loader keeps up to the top-10 axial slices by foreground area, requires at least 64 positive pixels, and uses a subject-level 80/20 split with NumPy seed 42. The $3{,}228$ count is traceable to the released `test_ids`: 321 held-out subjects contribute ten eligible slices, one contributes nine, one contributes six, and one contributes three, giving a 12-slice deficit from the $324{\times}10$ maximum. ACDC LV similarly evaluates 361 LV-positive slices nested under 20 held-out patients. The main table is therefore a case-level audit, but the released `subject_level/*.json` files include the patient/subject unit IDs linked to `test_ids` and two cluster-aware checks. **(i) Cluster bootstrap over the case-scale statistic.** Resampling patients/subjects with replacement and keeping all slices within each sampled unit gives ACDC $\Delta{=}0.321$ [0.269, 0.374] and BraTS foreground $\Delta{=}0.315$ [0.283, 0.351], so the case-scale gap remains positive under unit-level resampling. **(ii) Subject-aggregated statistic.** Averaging each member's per-slice Dice within patient/subject before computing Pearson gives smaller but still positive gaps: ACDC $\Delta{=}0.262$ [0.151, 0.413] and BraTS foreground $\Delta{=}0.214$ [0.182, 0.251]. The conservative reading is that slice dependence inflates the magnitude relative to subject means, especially on BraTS, but it does not explain the within-vs-cross separation away.

## A12 BraTS sparse-class filtering caveat and released diagnostics

Per-class BraTS sensitivity can be inflated by sparse labels: if a class is absent from a slice, empty-vs-empty Dice can return a perfect score and obscure the foreground-error structure that matters for a segmentation audit. For this reason, the primary BraTS row uses the foreground derivative `seg>0` rather than a sparse NCR-only target. NCR filtering diagnostics from a separate sparse-class sweep are not used as numeric evidence here because their exact aggregate files are not included in the supplementary artifact. The released BraTS foreground train-size and multimodal diagnostics are instead reported later in Appendices A31.5–A31.6 from `results/_merged/late_brats_summaries.json`.

## A13 Scope-only exploratory probes

*Note on CLIP-L on ACDC.* The secondary 8-FM ACDC diagnostic discussed in this section uses a different split where ResNet-50, ResNet-18, and CLIP ViT-L/14 collapse below the Dice$\geq$0.30 floor (Dice 0.099, 0.230, and 0.007 respectively); the *primary* 11-FM ACDC LV pool of Table 1 retains CLIP ViT-L/14 because it passes the Dice floor on the primary split. Different splits, different functional outcomes for the borderline bundles.

This block separates released diagnostics from exploratory mechanism probes without released trace artifacts or aggregate JSONs. Only released traces or released aggregate JSONs are used as numerical support for the paper's claims. The shared reading is descriptive: same-backbone perturbations can reduce correlation, but the released diagnostics do not make same-backbone pools behave like the primary cross-family references.

**ACDC architecture caveat.** Secondary ACDC exploratory tables used a separate 8-bundle split and architecture-pair subsets that are not included as replayable aggregate artifacts. The released support for the architecture-caveat wording is instead the same-architecture-family ViT / different-pretraining diagnostic in Appendix A29 and the expanded same-family cross-checkpoint decomposition in Appendix A30.

**Random-init probes.** DINOv2-B random-initialisation probes used secondary splits and source prediction caches outside the released artifact. The released no-pretraining evidence is limited to the CNN random-init diagnostic in Appendix A31.2, which should be read as a descriptive same-recipe sanity check rather than a causal isolation of architecture, optimiser, and data effects.

**LoRA and cross-architecture probes.** LoRA probes and ViT-B cross-architecture random-init sweeps require source prediction caches and aggregate summary files outside the released artifact. We do not tabulate their values or use them as bundle-supported evidence. The released same-backbone levers are the head-capacity sweep, UNet-skip replication, fold-reslicing diagnostic, CNN random-init diagnostic, and MC dropout.

## A14 Same-backbone, different-data bootstrap-bagging audit

For each (FM, dataset) pair, four bootstrap replicates of the training set (image-level on RIGA/Kvasir; subject-level on BraTS) are crossed with four decoder seeds. Bootstrap data-swap reduces within-pool $\bar{r}$ by only 0.05–0.12, while backbone swap reduces it by 0.33–0.50. This supports the scoped claim that, in the frozen-feature protocol, resampling the training cases does not provide the same decorrelation as changing encoder family.

## A15 Distribution-shift audit

We evaluate one released cross-distribution robustness diagnostic for the within–cross correlation gap using an 8-bundle RIGA pool and the frozen-feature protocol from Section 4.

**RIGA leave-MESSIDOR-out (cross-vendor fundus).** RIGA+ [Almazroa, 2018] combines three imaging sources: *BinRushed* (Bin Rushed Ophthalmic Center, Riyadh, Saudi Arabia), *Magrabia* (Magrabi Eye Center, Riyadh, Saudi Arabia), and *MESSIDOR* (French public corpus, different cameras and populations). We train the frozen-FM + $1\times1$ conv decoder on BinRushed+Magrabia

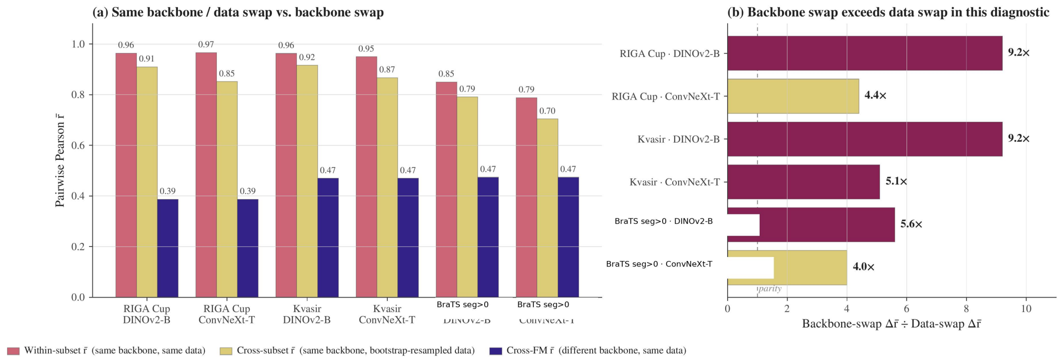


Figure 2: **Same-backbone bootstrap-bagging diagnostic.** For each shown dataset/backbone pair, we compare (i) same-backbone/same-data cross-seed correlation, (ii) same-backbone/bootstrap-resampled-data correlation, and (iii) a cross-backbone reference from the corresponding diagnostic pool. The ratio panel reports $(r_{\text{same data}} - r_{\text{cross backbone}})/(r_{\text{same data}} - r_{\text{bootstrap data}})$, so values above one mean that changing backbone reduces correlation more than bootstrap-resampling the training data in this diagnostic. The BraTS label refers to the 2D foreground derivative (`seg>0`), not the official 3D BraTS WT challenge target. This figure is descriptive and does not identify a causal mechanism.

(union) and evaluate on MESSIDOR ($454$ cases after the preprocessing and eligibility filters implemented by the released scorer) as cross-vendor out-of-distribution. Per-case traces for this diagnostic are released under `results/_merged/per_case_dice/riga_cup_ood_messidor`. The released scorer uses a fixed 8-bundle $\times$ 4-seed pool without a functional-floor change between ID and OOD rows. Under that directly reproducible convention, the ID row has within/cross/gap $0.659/0.282/0.377$ and the MESSIDOR OOD row has $0.743/0.390/0.353$.

**Pool-level summary.** In this fixed-pool RIGA cross-vendor diagnostic, the within–cross gap changes by $-0.024$ and remains positive. Both within and cross correlations are higher on the MESSIDOR row, so this diagnostic should be read only as evidence that the gap survives one released cross-vendor stress test, not as evidence that distribution shift generally contracts dependence. Broader cross-scanner, cross-demographic, and longitudinal shifts remain out of scope.

Table 15: **RIGA cross-vendor diagnostic under a fixed 8-bundle scorer.** This is not case-identical to Table 1. Within = same-FM different-seed $\cup$ same-family different-FM (DINOv2-B/S and ResNet-18/50); cross = different-family different-FM. The gap remains positive, but both within and cross correlations increase on MESSIDOR; no general distribution-shift conclusion is supported. Brackets are case-level bootstrap $95\%$ CIs ($B{=}500$). Source: `results/_merged/diagnostics/distshift_riga_messidor.json`.

| Shift regime | Within-$\bar{r}$ | Cross-$\bar{r}$ | Gap |
|---|---|---|---|
| ID reference (fixed 8-bundle pool) | $0.659$ [$0.603$, $0.715$] | $0.282$ [$0.197$, $0.369$] | $0.377$ [$0.301$, $0.463$] |
| OOD MESSIDOR (fixed 8-bundle pool) | $0.743$ [$0.722$, $0.765$] | $0.390$ [$0.345$, $0.430$] | $0.353$ [$0.318$, $0.388$] |

This diagnostic uses a paired fixed 8-bundle RIGA pool and should not be read as a case-identical comparison to the Magrabia-source held-out primary RIGA rows in Table 1.

## A16 UNet-with-skip decoder replication

The paper's primary decoder is a $1{\times}1$ conv with $C{+}1$ trainable parameters for $C$ frozen channels (for example, 769 parameters for a 768-channel feature map), which is close to a linear probe on frozen features. Dense decoders with skip connections—nnU-Net, U-Net, SegFormer—are closer to deployed medical segmentation. We therefore ran a separate diagnostic in which the frozen backbone is unchanged but the head is replaced by a multi-scale UNet-style decoder over four encoder feature levels. This is a *two-seed diagnostic grid* (seeds 42/43), not an inferential table with the same status as the four-seed primary pool; the reported correlation is the single seed-pair Pearson coefficient for each cell.

**Architecture.** For hierarchical CNNs (ConvNeXt-T, ResNet-50) the decoder uses the native feature pyramid as skip sources; for ViT-style encoders (DINOv2-B/S and BiomedCLIP ViT) it uses sampled intermediate token maps reassembled into a four-level pyramid. The released head first applies lateral $1\times1$ projections to a common decoder width, then runs a top-down bilinear-upsample path with three additive skip merges (deep $\rightarrow$ mid $\rightarrow$ shallow $\rightarrow$ shallowest), two conv-BN-ReLU blocks after each merge, a final conv-BN-ReLU after upsampling to the output grid, and a final $1\times1$ output conv. Training matches the frozen-feature recipe where possible: Adam lr $10^{-3}$, batch 16, 30 epochs, BCE, seeds $\{42, 43\}$.

Table 16: **Two-seed UNet-skip same-backbone diagnostic.** Each cell uses the two released seeds 42/43 from the UNet-skip trace set; "seed-pair $r$" is therefore one Pearson coefficient, not a bootstrapped four-seed within-family estimate. Primary Table 1 cross references are Kvasir 0.697, ACDC 0.639, RIGA Cup 0.361, and BraTS foreground 0.623.

| Task | Backbone | Mean Dice | Seed-pair $r$ |
|---|---|---|---|
| Kvasir | DINOv2-B | 0.883 | 0.877 |
| Kvasir | DINOv2-S | 0.866 | 0.919 |
| Kvasir | BiomedCLIP | 0.806 | 0.926 |
| Kvasir | ConvNeXt-T | 0.860 | 0.775 |
| Kvasir | ResNet-50 | 0.816 | 0.774 |
| ACDC LV | DINOv2-B | 0.864 | 0.979 |
| ACDC LV | DINOv2-S | 0.863 | 0.970 |
| ACDC LV | BiomedCLIP | 0.827 | 0.950 |
| ACDC LV | ConvNeXt-T | 0.885 | 0.917 |
| ACDC LV | ResNet-50 | 0.882 | 0.945 |
| RIGA Cup | DINOv2-B | 0.902 | 0.967 |
| RIGA Cup | DINOv2-S | 0.894 | 0.893 |
| RIGA Cup | BiomedCLIP | 0.880 | 0.892 |
| RIGA Cup | ConvNeXt-T | 0.895 | 0.970 |
| RIGA Cup | ResNet-50 | 0.881 | 0.952 |
| BraTS foreground | DINOv2-B | 0.877 | 0.936 |
| BraTS foreground | DINOv2-S | 0.875 | 0.886 |
| BraTS foreground | BiomedCLIP | 0.858 | 0.929 |
| BraTS foreground | ConvNeXt-T | 0.895 | 0.938 |
| BraTS foreground | ResNet-50 | 0.889 | 0.934 |

**Finding.** The UNet-skip head improves mean Dice substantially over several weak $1\times1$ probe cells, but it does not remove same-backbone co-failure: across the 20 evaluated cells, the seed-pair correlation remains $0.774$–$0.979$. On rows where the matched $1\times1$ seeds 42/43 are both available, UNet-skip usually lowers seed-pair $r$ modestly, but the effect is heterogeneous: Kvasir ResNet-50 drops $0.973\rightarrow0.774$, while RIGA Cup DINOv2-B is flat ($0.965\rightarrow0.967$) and RIGA Cup ResNet-50 increases ($0.908\rightarrow0.952$). The BraTS cells are consistent with the same boundary: DINOv2-B/S UNet-skip reaches mean Dice $0.875$–$0.877$, but seed-pair $r$ remains $0.886$–$0.936$. Thus the conclusion is narrower than a decoder-family claim: a stronger skip decoder can improve accuracy and sometimes reduce seed correlation, but under this frozen-backbone two-seed grid it does not make same-backbone pools resemble the cross-family references in Table 1. A full nnU-Net v2 5-fold audit on RIGA Cup is reported in Appendix A17.

## A17 Fold-reslicing and nnU-Net complements

**Motivation.** nnU-Net [Isensee et al., 2021] is the de facto standard for medical image segmentation; its 5-fold cross-validation ensemble is a routinely-used uncertainty / robustness baseline. The full nnU-Net pipeline is *self-configuring* (2D/3D selection, preprocessing, patch-based sampling, default geometric/intensity augmentation, cascaded network, post-processing). We keep three evidence surfaces separate: (i) the released frozen-feature fold-reslicing diagnostic, which tests split sensitivity with one decoder seed per fold and therefore cannot estimate within-fold versus cross-fold same-backbone pairs; (ii) released task-level nnU-Net summary JSONs, which are full-pipeline complements but not case-identical contrasts to the primary FM rows; and (iii) a single case-identical RIGA Cup full-pipeline scope check on the primary Magrabia held-out cases.

**Released fold-reslicing design.** This diagnostic re-slices the frozen-feature cache into 5 folds and reruns the 8-bundle pool {DINOv2-B/S, BiomedCLIP, CLIP-L, ConvNeXt-T, EfficientNet-B0, ResNet-50/18} on ACDC LV and RIGA Cup, single seed per fold cell. Because the released artifact contains one seed per fold, this diagnostic reports per-fold cross-bundle $\bar{r}$ and pool-mean Dice, not a 20-decoder fixed-backbone within/cross estimator. The released artifact provides the corresponding fold-reslicing summary and scorer.

Table 17: **Fold-reslicing cross-bundle stability, one seed/fold.** 8 bundles $\times$ 5 folds; single seed per fold cell. Cross-bundle $\bar{r}$ is the Pearson correlation averaged over the 28 bundle pairs on the held-out fold. This table cannot support within-fold versus cross-fold same-backbone claims.

| Task | fold 0 | fold 1 | fold 2 | fold 3 | fold 4 | mean $\pm$ std |
|---|---|---|---|---|---|---|
| ACDC LV | $0.580$ | $0.548$ | $0.583$ | $0.558$ | $0.558$ | $0.565 \pm 0.015$ |
| RIGA Cup | $0.359$ | $0.317$ | $0.325$ | $0.315$ | $0.261$ | $0.315 \pm 0.035$ |

**Finding.** Train/test split sensitivity is small in this fold-reslicing check: cross-fold pool-mean Dice is stable (ACDC $0.502 \pm 0.022$; Cup $0.610 \pm 0.012$), and per-fold cross-bundle $\bar{r}$ is stable across folds (ACDC $0.565 \pm 0.015$; Cup $0.315 \pm 0.035$). This supports only the narrower statement that the cross-bundle diagnostic is not dominated by one particular 5-fold re-slicing. It does *not* support a claim about within-fold vs cross-fold same-backbone decorrelation, because those pair types require multiple decoder seeds per fold that are not present in the released traces.

**Full nnU-Net v2 complement.** To complement the frozen-feature proxy above and answer whether nnU-Net's full self-configuring pipeline decorrelates the pool more, we trained the full nnU-Net v2 2D U-Net on RIGA Cup with a $520/224$ split, using the generated v2 2D `nnUNetPlans` configuration after `nnUNetv2_plan_and_preprocess` (self-configured preprocessing and architecture, official default augmentation/loss schedule, $250$ iterations per epoch). This split differs from the Magrabia-source held-out FM row and is therefore a task-level complement. We train two seeds per fold via custom `nnUNetTrainerSeed13_100epochs` and `nnUNetTrainerSeed37_100epochs` subclasses that pre-seed `random`, `np.random`, `torch`, and `cuda` before any nnU-Net sampler touches the RNG; each is trained for $100$ nnU-Net-epochs ($25{,}000$ iterations on two A100s in parallel), then all $5 \times 2 = 10$ models predict on the held-out $224$-image test set via `nnUNetv2_predict` and per-case Dice is computed against the held-out labels staged in `labelsTr`. A split-index file enumerated held-out IDs; the artifact bundle releases the summary arrays/JSONs for this complement but not the split manifest.

Across all $10$ models the mean Dice is $0.949 \pm 0.001$ (exceeding any of our frozen-feature FM+$1\times1$-conv combinations on RIGA Cup). The pairwise per-case Dice $\bar{r}$ is $0.972$ within-fold over the $5$ same-fold different-seed pairs and $0.925$ cross-fold over the $40$ different-fold any-seed pairs, with $\Delta_{\mathrm{within-cross}}=0.047$ and a case-level bootstrap $95\%$ CI of $[0.030, 0.069]$ that excludes zero. Full nnU-Net v2 is structurally a same-architecture, different-training-data ensemble on one 2D PlainConvUNet, so it is not comparable in kind to the primary frozen-FM cross-family reference, which swaps encoder bundles. The scoped conclusion is therefore: *within this task-level same-architecture nnU-Net complement, fold identity leaves a small but positive correlation gap*; we do not claim nnU-Net substitutes for or surpasses backbone-family diversification. The 100-epoch budget, single-task coverage, and 2D-only scope are closed in Appendix A18 (1000-epoch retrain on the same RIGA Cup setup; 3D fullres nnU-Net on ACDC; higher-resolution/Dice+BCE/aug frozen-FM replication).

**Case-identical RIGA Cup held-out scope check.** We additionally train the same $5 \times 2$ seeded nnU-Net v2 2D grid on BinRushed+MESSIDOR folds and evaluate all models on the primary Magrabia-source held-out Cup cases ($n$=95). The exporter uses the official nnU-Net v2 natural-image channel convention, with RGB saved as `_0000/_0001/_0002` channel files, `imagesTs/labelsTs` for held-out scoring, and the generated `nnUNetPlans` 2D configuration. Released case IDs are deterministic hashes rather than upstream path strings. Across the 10 models, mean Dice is $0.916 \pm 0.002$. The same-fold, different-seed correlation is $\bar{r}$=0.984 over 5 pairs, the different-fold correlation is $\bar{r}$=0.976 over 40 pairs, and the gap is $\Delta$=0.008 with case-bootstrap $95\%$ CI $[0.002, 0.039]$. This confirms the boundary in a stricter paired setting: nnU-Net's same-architecture fold/seed surface is highly correlated on the same held-out cases, and the fold gap is much smaller than both the primary RIGA Cup same/cross gap ($0.473$) and the separate-split task-level nnU-Net summary ($0.047$). It does not make nnU-Net a cross-family reference or a causal control for architecture versus pretraining.

**Data-swap comparison caveat.** Appendix A14 reports bootstrap-bagging within-pool decorrelation with four decoder seeds per bootstrap replicate. The fold-reslicing row instead reports single-seed cross-bundle correlations across held-out folds. These diagnostics both stress training data/split sensitivity, but they are not dose-matched estimators and are not combined into a causal ranking.

## A18 Segmentation-recipe and nnU-Net scope audit

Our main protocol uses $224\times224$ 2D inputs, frozen backbone, $1\times1$-conv decoder, BCE loss, no augmentation, and 30-epoch head training. Routine medical-imaging pipelines often run with compound Dice+CE loss, geometric augmentation, 3D input, and full-length (1000-epoch) training, so we test whether the within/cross correlation structure we report is specific to the simplified protocol. Most nnU-Net rows below are task-level complements; the added case-identical RIGA held-out row uses the same Magrabia-source held-out cases as the primary frozen-FM row but remains a single-task, 2D, same-architecture nnU-Net scope check.

**Dimensionality axis: 3D fullres nnU-Net at 1000 epochs on ACDC.** Training nnU-Net v2 in its self-configuring `3d_fullres` setting on ACDC with a seed-42 $105/45$-patient split (210 training, 90 test volumes) using `nnUNetTrainerSeed{13,37}_1000epochs` (seed injected in `on_train_start` before sampler initialisation) gives 10 fully-trained 3D models across 5 folds. `nnUNetv2_predict` on all 90 held-out volumes yields per-case Dice on mean foreground and per class. Across 10 models the mean Dice is $0.920 \pm 0.001$, within the range reported for strong nnU-Net baselines on ACDC. On per-case mean Dice the within-fold (5 pairs) vs cross-fold (40 pairs) gap is $\Delta$=$0.043$ [0.021, 0.085] with within $\bar{r}$=$0.978$ and cross $\bar{r}$=$0.935$; per class, $\Delta_{\mathrm{RV}}$=$0.041$ [0.011, 0.110], $\Delta_{\mathrm{Myo}}$=$0.029$ [0.017, 0.046], $\Delta_{\mathrm{LV}}$=$0.067$ [0.033, 0.098]. The artifact bundle releases the aggregate summary for this 150-patient complement but not the exact split manifest; the released public ACDC converter targets patients 001–100 for the primary 2D row. We therefore treat this 3D row as a task-level scope summary rather than a from-bundle training recipe. In this ACDC nnU-Net complement, the 2D$\rightarrow$3D fullres shift changes the small fold-gap regime by less than $0.01$ relative to the 2D 100-epoch baseline ($\Delta$=$0.047$ in Appendix A17).

**Loss / augmentation axis: RIGA Cup frozen-FM at encoder-native $224\times224$.** We retrain the seven-bundle RIGA Cup subset (DINOv2-B/S, CLIP ViT-L/14, ConvNeXt-T, EfficientNet-B0, ResNet-50/18) with four seeds each under a more standard segmentation loss/augmentation recipe: encoder-native input size $224\times224$, composite soft Dice+BCE loss, paired random horizontal flip and rotation $\pm15°$, for 30 epochs. The frozen encoder wrappers use $224\times224$ inputs; we therefore treat this as a loss/augmentation test, not a resolution test. All other factors (frozen backbone, $1\times1$-conv segmentation head, Adam $10^{-3}$) match the main protocol. Per-FM mean Dice ranges from $0.526$ (ResNet-18) to $0.758$ (DINOv2-B). Under the same symmetric estimator used for the main table, within-family $\bar{r}$ is $0.903$ (58 within pairs, including same-FM seed pairs and DINOv2-B$\times$S), cross-FM $\bar{r}$ is $0.556$ (320 cross pairs), and the within/cross gap is $\Delta$=$0.347$ [0.275, 0.428]. The gap is smaller than the main RIGA Cup interval ($0.473$ [0.395, 0.556]) but remains positive; loss/augmentation changes do not remove the gap.

**Training-length axis: full nnU-Net v2 at the 1000-epoch default on RIGA Cup.** We retrain the 10-model RIGA Cup 2D nnU-Net pool from Appendix A17 at the nnU-Net v2 default 1000 epochs, using the same seeded trainer pattern (`nnUNetTrainerSeed{13,37}_1000epochs`) on the same $520/224$ split with identical dataset/plan files. Mean Dice across the 10 models is $0.946 \pm 0.0004$ (vs $0.949 \pm 0.001$ at 100 epochs). Within-fold $\bar{r}$=$0.983$ (5 pairs), cross-fold $\bar{r}$=$0.949$ (40 pairs), and $\Delta$=$0.034$ [0.021, 0.058]. Relative to the 100-epoch run both estimators tighten slightly (within $0.972\rightarrow0.983$; cross $0.925\rightarrow0.949$) and the within/cross gap contracts from $0.047$ to $0.034$; measured predictions are more correlated under longer training, and the gap does not disappear. Ten-fold longer training therefore does not change the replicability structure of nnU-Net on this task-level complement; it is not a case-identical comparison to the Magrabia-source held-out FM row.

**Synthesis.** Across these scope analyses, the frozen-FM loss/augmentation row remains in the same large-gap regime as the RIGA Cup main row, while the same-architecture nnU-Net fold axis remains in a small-gap regime. The case-identical RIGA held-out row tightens this boundary: on the same Magrabia cases, both nnU-Net correlations are very high and the fold gap is only $0.008$ at 100 epochs. These rows do not support a simpler explanation that the main-table pattern is solely an artifact of BCE loss, no augmentation, short training, or the tested full-pipeline nnU-Net complements.

**Pending case-identical 1000-epoch RIGA run.** A companion 1000-epoch RIGA *case-identical* run (Magrabia-source held-out test cases) is in progress; on completion the row will be inserted into Table 18 following the procedure documented in `reports/12_pending_1000ep_insertion_points.md`. The current 1000-epoch RIGA Cup row in this section uses the separate $520/224$ task-level split, not the Magrabia-source held-out split.

Table 18: **Scope audit across recipe and nnU-Net complements.** Rows are not all case-identical and should not be compared as a controlled ablation. The table asks whether selected recipe changes remove the qualitative gap. The frozen-reference row uses the same seven candidate FMs as the Dice+BCE+augmentation row before the primary functional-floor exclusion; the primary filtered RIGA Cup row is Table 1. The case-identical RIGA held-out row uses the Magrabia-source test cases; the separate-split RIGA nnU-Net rows use a 520/224 split.

| Axis | Protocol | Task | Within | Cross | $\Delta$ | 95% CI |
|---|---|---|---|---|---|---|
| Frozen reference | 7-bundle candidate set, 224, BCE, no aug, 30 epochs | RIGA Cup | $0.798$ | $0.266$ | $0.532$ | $[0.456, 0.614]$ |
| Loss / aug | 7-bundle frozen, 224, Dice+BCE, aug, 30 epochs | RIGA Cup | $0.903$ | $0.556$ | $0.347$ | $[0.275, 0.428]$ |
| Baseline (App. A17) | nnU-Net 2D, 100 epochs, separate split | RIGA Cup | $0.972$ | $0.925$ | $0.047$ | $[0.030, 0.069]$ |
| Case-identical held-out | nnU-Net 2D, 100 epochs, Magrabia test | RIGA Cup | $0.984$ | $0.976$ | $0.008$ | $[0.002, 0.039]$ |
| Training length | nnU-Net 2D, 1000 epochs, separate split | RIGA Cup | $0.983$ | $0.949$ | $0.034$ | $[0.021, 0.058]$ |
| Dimensionality | nnU-Net 3D fullres, 1000ep | ACDC (mean) | $0.978$ | $0.935$ | $0.043$ | $[0.021, 0.085]$ |
| | per-class RV | | $0.967$ | $0.927$ | $0.041$ | $[0.011, 0.110]$ |
| | per-class Myo | | $0.987$ | $0.957$ | $0.029$ | $[0.017, 0.046]$ |
| | per-class LV | | $0.978$ | $0.911$ | $0.067$ | $[0.033, 0.098]$ |

## A19 Held-out multi-task stress test

**Summary.** Across the five held-out 2D binary derivatives we tested, two rows give substantively positive gaps (Hippocampus $\Delta$=$0.491$, ISIC-2018 $\Delta$=$0.406$), one is tiny-$N$ positive (Prostate, $n$=6, $\Delta$=$0.146$), one gives a near-zero gap (Spleen $\Delta$=$0.003$), and one fails the functional floor (Heart, $0/11$ functional FMs). The held-out evidence *supports* rather than confirms the primary 2D rows and is consistent with task-specific structure modulating the gap magnitude.

**Protocol.** In addition to the four primary benchmark datasets (five task rows: Cup, Disc, ACDC, Kvasir, BraTS foreground), we evaluate five *held-out* tasks covering additional imaging domains: MSD Hippocampus [Antonelli et al., 2022] (T1 MRI, native 3-class mask {background, anterior, posterior} binarized to {background, any-hippocampus}), ISIC-2018 Task 1 Lesion Boundary [Codella et al., 2019] (dermoscopy, 2D RGB, $N$=80 test cases held out from a 400-image random subset of the 2,594-image training release), MSD Heart (T2 MRI cardiac), MSD Prostate (T2 MRI prostate), and MSD Spleen (CT portal). This held-out stress test uses the same 11 generic encoder bundles and four decoder seeds as the primary source pool where possible, with the same $1\times1$-conv decoder, no per-task hyperparameter tuning, and no model substitution. These are task-specific 2D binary derivatives, not official challenge-test evaluations. The Dice$\geq$0.30 functional floor is applied uniformly, and the released aggregate summary records 220/220 expected per-seed traces.

**Interpretation.** The 11-bundle held-out protocol yields three classes of outcome: (i) ISIC-2018 and Hippocampus both show direction-positive gaps under the same broad estimator family as the primary audit, but without case-bootstrap CIs and with smaller held-out support than the primary rows. (ii) Prostate and Spleen illustrate why held-out diagnostics are scope boundaries rather than primary evidence: Prostate is direction-positive but has only six test cases, while Spleen passes the floor yet has a near-zero gap and $M_{\text{eff}} \approx 1$ because the members fail together. (iii) Heart does not yield a functional pool under the frozen-feature 2D derivative. Thus the held-out stress test is compatible with transfer to some external rows, but it also records where the protocol is too weak or too saturated to audit same-bundle dependence.

**Operational 8-bundle held-out probes.** The following recovery, pool-size, abstention, and pool-selection diagnostics use an 8-bundle held-out candidate set stored under `results/_merged/diagnostics/held_out/`. They are retained as operational-method sensitivities, not as the provenance for Table 19 above.

**CLIP/BiomedCLIP normalization robustness.** Uniform ImageNet normalization can be a potential artifact for BiomedCLIP and CLIP-L because both have their own normalization statistics. We therefore evaluated those two FMs on Hippocampus and ISIC-2018 with the correct CLIP nor-

Table 19: **Held-out aggregate stress test** (11 generic encoder bundles $\times$ 4 decoder seeds; point estimates only, no case-bootstrap CIs). Hippocampus and ISIC-2018 are direction-positive, Prostate is positive but has only six test cases, Spleen is essentially near-zero despite passing the Dice floor, and Heart has no functional bundle at the Dice$\geq$0.30 floor. Tiny-$N$ and floor-failure rows are scope boundaries and do not support the primary claim.

| Task | $N$ | Func./avail. | Within | Cross | $\Delta$ | $M_{\text{eff}}$ |
|---|---|---|---|---|---|---|
| *Direction-positive point estimates with enough support to read qualitatively* | | | | | | |
| MSD Hippocampus | 52 | 11/11 | 0.909 | 0.419 | **0.491**[§] | 3.94 |
| ISIC-2018[‡] | 80 | 11/11 | 0.918 | 0.511 | **0.406**[§] | 2.72 |
| *Scope-limit rows: tiny held-out support or non-functional pool* | | | | | | |
| MSD Prostate | 6 | 5/11 | 0.788 | 0.642 | 0.146[§] | 1.23 |
| MSD Spleen | 8 | 11/11 | 0.983 | 0.979 | 0.003[§] | 1.02 |
| MSD Heart | 4 | 0/11 | —[*] | —[*] | —[*] | — |

[*]No bundle passes the Dice$\geq$0.30 floor on the four-case Heart derivative, so no within/cross gap is interpretable. [§]Values are point estimates from `results/_merged/diagnostics/heldout_11fm_summary.json`; case-level CIs are not computed for this aggregate diagnostic, and the MSD Prostate/Spleen rows have too few held-out cases to support a primary claim. [‡]ISIC-2018 uses a single random $N$=400-image subset of the 2,594-image ISIC-2018 training release with $N$=80 test cases under the seed-42 80/20 split; we do not report an independent redraw and we do not use the official ISIC test server in this study.

malization. The released `summary_clipfix.json` reports BiomedCLIP mean Dice $0.840$ (Hippocampus) / $0.908$ (ISIC-2018), while CLIP-L remains below the functional floor on Hippocampus (mean Dice $<10^{-10}$) and reaches $0.106$ on ISIC-2018, still below the $0.30$ floor. Relative to `summary_5tasks.json`, the aggregate $\Delta$ changed by $0.002$ (Hippocampus $0.731 \rightarrow 0.729$) and $0.003$ (ISIC-2018 $0.552 \rightarrow 0.555$). In this held-out diagnostic, correcting CLIP/BiomedCLIP normalization does not materially change the Hippocampus/ISIC point estimates.

**Held-out recovery-rate sensitivity.** To complement the primary-task $\Delta_{\text{rec}} = +\,0.29$ (Cup) / $+0.26$ (ACDC) grand means, we computed an exploratory held-out recovery gap at $\tau$=0.85 on the single ISIC-2018 subset: $\Delta_{\text{rec}} = +\,0.269$ [+0.212,+0.331] under a case-resampling calculation saved as `recovery_CI.json`. This supports the direction of the operational gap on one dermoscopy subset, but it is not a deployment-calibrated guarantee and does not resolve the held-out estimator-dependence above. Hippocampus at $\tau$=0.85 is recovery-saturated and gives $\Delta_{\text{rec}} = -\,0.01$ [$-0.03$, $+0.00$] (diverse and mono both near-zero recovery because only 2/7 functional bundles individually pass Dice$\geq$0.85). Lower thresholds are not evaluated here.

**Pool-size sweep on held-out tasks.** How does diverse-vs-same-bundle conditional recovery scale with $M$? For $M \in \{2, 3, 4, 5, 6\}$ on the two functional held-out tasks (Table 20), diverse pools beat same-bundle seed pools at every $M \geq 3$ on the single ISIC-2018 subset ($\Delta_{\text{rec}}$=0.27–0.46), while Hippocampus at the paper's $\tau$=0.85 threshold is recovery-rate-saturated (all pools $<$0.11) because only 2 of 7 functional bundles reach Dice$\geq$0.85 individually. These held-out operational analyses are exploratory: ISIC provides one supportive dermoscopy subset, Hippocampus is threshold-saturated at $\tau$=0.85, and the $M$=5, 6 same-bundle rows are proxy constructions using the available four seeds rather than true larger same-bundle pools. Released aggregate data: `results/_merged/diagnostics/held_out/pool_size_sweep.json`.

**Calibration-tuned disagreement-based abstention (an exploratory alternative lever, not a conformal guarantee).** A natural operational question is whether standard abstention-based uncertainty gating closes the same-bundle gap with modest additional computation. We calibrate a per-task disagreement-score threshold on 20 cases (inter-member Dice standard deviation as score; grid-search the quantile $q$ to meet a target case-risk $\alpha$=0.1 on calibration; apply the selected threshold to the remaining test cases). *This is a tuned heuristic, not split-conformal prediction* [Angelopoulos and Bates, 2021]: we use a grid search over $q$ that terminates on the first calibration risk meeting $\alpha$ and use inter-member variance rather than a proper conformity score, so no finite-sample risk guarantee holds. The pool is also chosen per-task on the whole task mean Dice (family-representative highest-Dice FMs), so the diverse-pool selection step leaks task labels—this biases the comparison toward diverse pools. We report it as a sensitivity. On ISIC-2018 at matched abstention $\approx$ 0.74, diverse pools reach test-risk 0.129±0.027 vs. same-bundle grand mean 0.174±0.209 (`conformal_vs_meff.json`). The *point estimate* is $\sim$ 25% lower for diverse, but the same-bundle grand-mean SD (0.21) is $\sim$ 8$\times$ the diverse SD (0.027); the per-FM same-bundle risk means span 0.000–0.616 across the seven functional FMs, so the grand-mean comparison is highly composition-sensitive and we do not claim a tight ranking. Hippocampus

Table 20: **Exploratory held-out pool-size sweep.** Conditional recovery rate $P$[ens pass | any fail] at $\tau$=0.85 vs pool size $M$. $M$=2 recovery is a definition artifact under the strict-majority rule, and $M$=5, 6 same-bundle values are four-seed proxies. This table is not a deployable pool-size recommendation.

| | Hippocampus | | | ISIC-2018 | | |
|---|---|---|---|---|---|---|
| $M$ | mono | diverse | $\Delta$ | mono | diverse | $\Delta$ |
| $2^{\dagger}$ | 0.000 | 0.000 | +0.00 | 0.000 | 0.000 | +0.00 |
| 3 | 0.069 | 0.107 | +0.04 | 0.111 | 0.498 | +0.39 |
| 4 | 0.052 | 0.039 | −0.01 | 0.091 | 0.359 | +0.27 |
| $5^{\ddagger}$ | 0.052 | 0.078 | +0.03 | 0.091 | 0.553 | +0.46 |
| $6^{\ddagger}$ | 0.052 | 0.027 | −0.03 | 0.091 | 0.454 | +0.36 |

$^{\dagger}$ $M$=2 is a definition artifact under the strict-majority ($>M/2$) rule conditioned on any-fail: recovery is zero by construction whenever $M$=2, regardless of pool composition. We include the row for completeness. $^{\ddagger}$ At $M$=5, 6 the same-bundle entries are computed on the $\min(M, S)$=4 available seeds per FM, so they do not represent true $M$=5, 6 same-bundle pools; the diverse entries do use legitimate $M$=5, 6 distinct-family compositions.

cannot hit $\alpha$=0.1 for either pool, giving residual risk 0.694 (diverse) vs. 0.665 (same-bundle grand mean) ($\Delta = +\,0.03$, with same-bundle SD 0.28, indistinguishable). A label-free conformal procedure with a proper conformity score is not evaluated here. Released aggregate data: `results/_merged/diagnostics/held_out/conformal_vs_meff.json`.

**Calibration-set pool-selection probe on held-out tasks.** We operationalise a small calibration exercise: given only a 20-case calibration set on a held-out task, can a rule pick a 4-of-8 FM pool from the same secondary held-out candidate set that beats random? We evaluate five rules—(R1) random, (R2) top-4 by mean Dice on calibration, (R3) max-$M_{\text{eff}}$, (R4) quality-gated max-$M_{\text{eff}}$ (Dice$\geq$0.30 floor then max-$M_{\text{eff}}$ among qualifying), (R5) diverse-by-architecture (2 best-Dice ViT + 2 best-Dice CNN; reported as an exploratory hand-crafted baseline, not a general proposal)—and report conditional recovery rate $P$[ens pass | any fail] at $\tau$=0.85 on the remaining held-out cases, averaged over 100 calibration-set bootstraps (Table 21). All rule selectors and the recovery estimator use seed-42 per-case Dice (not seed-averaged), matching a one-model-per-FM pool construction. Hippocampus has lower effective recovery support than ISIC at $\tau$=0.85, so its recovery numbers are exploratory. In this diagnostic, simple top-4-by-Dice has the highest mean on both tasks (Hippocampus 0.161 vs. 0.043 random; ISIC-2018 0.581 vs. 0.378 random), while max-$M_{\text{eff}}$ alone is worse than random (0.011 / 0.327). This supports the narrower point that $M_{\text{eff}}$ is a dependence readout rather than a standalone operational objective; it is not a validated clinical pool-selection rule. Released aggregate data: `results/_merged/diagnostics/held_out/pool_selection_challenge.json`.

Table 21: **Exploratory pool-selection probe.** Conditional recovery rate at $\tau$=0.85 (mean ± SD over 100 calibration bootstraps, 20 cases each, using seed-42 traces). Selection uses calibration labels. This probes whether $M_{\text{eff}}$ alone is a poor objective; it is not a validated clinical selection rule.

| Rule | Hippocampus | ISIC-2018 |
|---|---|---|
| (R1) Random 4-of-8 | $0.043 \pm 0.050$ | $0.378 \pm 0.125$ |
| (R2) Top-4 by calibration Dice | $\mathbf{0.161} \pm 0.017$ | $\mathbf{0.581} \pm 0.030$ |
| (R3) Max $M_{\text{eff}}$ | $0.011 \pm 0.011$ | $0.327 \pm 0.100$ |
| (R4) Quality-gated max $M_{\text{eff}}$ | $0.016 \pm 0.014$ | $0.381 \pm 0.112$ |
| (R5) Diverse-by-arch (2 ViT + 2 CNN) | $0.096 \pm 0.017$ | $0.445 \pm 0.036$ |

**Held-out estimator note.** Split-stability and alternate-estimator numerical summaries without released aggregate files are omitted from the numeric evidence. This section therefore relies on the released 11-bundle held-out aggregate and the released operational JSONs listed above.

**ISIC-2018 single-subset note.** The ISIC-2018 row above is a single random $N$=400-image subset of the ISIC-2018 Task 1 training release with $N$=80 held-out test cases under a seed-42 80/20 split. We do not report an independent replication of this subset; a fully random redraw of the subset or use of the official ISIC evaluation server would be a natural next step but is out of scope for this paper.

## A20 Full fine-tuning scope

RIGA Cup full fine-tuning epoch-sweep traces and aggregate summaries are outside the released artifact. We therefore do not tabulate those values or use them as numerical evidence. The released adaptation evidence is limited to released traces and summaries named elsewhere in the appendix, including a small set of ACDC DINOv2-B full-fine-tuning provenance traces but not the RIGA epoch-sweep matrix; full-fine-tuning numbers not backed by released paths should be read only as exploratory mechanism notes and not as support for the paper's claims.

## A21 Same-backbone and scope diagnostic index

Table 22: **Qualitative index of same-backbone and scope diagnostics**. Rows are heterogeneous and not CI-comparable; the table is a navigation aid, not an ordered causal hierarchy. The RIGA-oriented map is anchored on the DINOv2-B 4-seed baseline ($\bar{r}$≈0.96) unless marked. LoRA, full-fine-tuning epoch-sweep, and ViT random-init probes without released trace artifacts are omitted.

| Diagnostic | Readout | Approx. change from stated baseline | Evidence |
|---|---|---|---|
| Mono baseline (same FM, 4 seeds) | 0.96 | 0.00 | RIGA diagnostic baseline |
| UNet-skip decoder (5 backbones, 4 tasks) | 0.774–0.979 | heterogeneous | Appendix A16, Tab. 16 |
| Fold-reslicing cross-bundle check | 0.315–0.565 | split-stable | Appendix A17; released summary |
| MC dropout $p$=0.1–0.4 | 0.89–0.95 | $-0.02$ to $-0.10$ | Appendix A31.7 |
| Same-backbone bootstrap-bagging | 0.85–0.92 | $-0.05$ to $-0.12$ | Fig. 2 |
| RIGA cross-vendor shift | OOD w/c $0.743/0.390$ | gap 0.353; $\Delta_{\text{gap}}-0.024$ | Appendix A15 |
| Decoder capacity sweep | $\sim 0.69$ | $\sim -0.27$ | Appendix A31.1 |
| **Cross-family pretrained 8-bundle pool** | $\mathbf{0.38}$ [0.29, 0.46] | $\mathbf{-0.58}$ | RIGA mechanism diagnostic |

## A22 Leave-one-family-out scope

Leave-one-family-out summaries were used as exploratory robustness checks, but the exact aggregate source is outside the released artifact. We therefore omit the numeric table from the released evidence record. The primary robustness check for the primary rows is the released `diagnostics/audit_sensitivity.json`, which includes leave-one-task and leave-one-FM summaries under the same scorer path.

## A23 Joint-failure lift at support limits

Joint-failure lift $L{=}P_{\text{empirical joint}}/\prod_m P_{\text{marginal}}$ is unstable when marginal fail rates are near 0 or near 1. When $P_{\text{marginal}} \approx 0.3$, independence-implied joint $\approx 0.01$; small sampling variation in the empirical joint produces large lift ratios. Representative near-ceiling cells: Only the ACDC 2.8× row is adequately supported by the $N \cdot \mathbb{E}[\mathrm{AF}_{\mathrm{ind}}] \geq 5$ rule. The RIGA Cup 6.1× and 38.8× rows are support-limited, and the 123.2× and 189× rows are extreme sparse-tail diagnostics.

Table 23: **Support-limit examples for joint-failure lift.** Extreme lifts are shown to explain instability, not to support a primary claim. Lifts $\geq$ 50× are descriptive (support-limited, wide noise envelope) and are *not* primary evidence.

| Task / Pool | $\tau$ | Indep. joint | Empirical joint | Lift |
|---|---|---|---|---|
| RIGA Cup mono BiomedCLIP | 0.85 | 0.0010 | 0.125 | 123.2× |
| RIGA Cup mono DINOv2-B | 0.85 | 0.0047 | 0.183 | 38.8× |
| RIGA Cup diverse 4-bundle | 0.85 | 0.0160 | 0.098 | 6.1× |
| ACDC LV mono DINOv2-B | 0.70 | 0.0002 | 0.038 | 189× |
| ACDC LV diverse 4-bundle | 0.85 | 0.135 | 0.378 | 2.8× |

## A24 3D segmentation-architecture pilot

This section provides scope-boundary context. Its per-case outputs and scorer version are not included in the supplementary artifact, so the pilot is not used as primary evidence for Table 1, the abstract, or the released artifact claims. It motivates why we avoid generalizing the frozen 2D FM-pool result to 3D segmentation-architecture pools.

We assembled a 3D segmentation-architecture pool for BraTS 2024 3D volumes (271 subjects, 216/55 train/test, seed 42) using three retained architectures: SegResNet (optionally initialised from the MONAI BraTS bundle), DynUNet (random-init), and UNETR (random-init). SwinUNETR random-init cells were excluded because predictions were degenerate across all seeds. For each retained (architecture, seed $\in \{42, 43, 44, 45\}$) cell, the pilot trained the full MONAI segmentation model for 100 epochs with Adam $10^{-3}$ and BCE loss; UNETR-style models used sliding-window inference where required by memory. Reported at three filter modes:

Table 24: **3D segmentation-architecture pilot.** Per-case outputs and scorer version are not included in the supplementary artifact; this table is excluded from the abstract and main-result evidence. Three functional-filter modes are applied uniformly in the pilot. Strict filter (Dice $\geq 0.50$) retains only SegResNet pretrained and DynUNet random-init and yields $\Delta$=$0.045$, substantially below the 2D BraTS foreground $\Delta$=$0.315$.

| Filter | FMs retained | Within | Cross | $\Delta$ [95% CI] |
|---|---|---|---|---|
| Mixed (all cells) | 3 (12 cells) | 0.772 | 0.595 | 0.177 [0.117, 0.248] |
| Dice $\geq 0.30$ | 3 (12 cells, drop UNETR seed 43 degenerate) | 0.919 | 0.703 | 0.216 [0.134, 0.271] |
| Dice $\geq 0.50$ (strict functional) | 2 (SegResNet pretrained, DynUNet random) | 0.916 | 0.871 | **0.045** [0.015, 0.112] |
| Random-init only (DynUNet + UNETR) | 2 random-init | 0.724 | 0.479 | 0.245 [0.155, 0.351] |

**Interpretation.** Within the strict functional regime (only converged pretrained + converged random-init CNN-style 3D architectures), the 3D segmentation-architecture pilot shows $\Delta$=$0.045$—substantially below the 2D BraTS foreground $\Delta$=$0.315$. We treat this only as a scope warning: the 2D frozen-feature same-bundle dependence claim should not be extended to 3D segmentation-architecture pools without a released, case-identical 3D benchmark. Random-init 3D architectures cross-correlate *more* than pretrained 3D architectures (within random $0.724$ vs cross-architecture random $0.479$, $\Delta$=$0.245$), so architecture-family effects again appear large in this pilot. The degenerate SwinUNETR pilot cells limit architecture coverage in the 3D pilot; expanded 3D foundation-model evaluation (Models Genesis, Vista3D, Triad-MRI) is not evaluated here.

## A25 Matched-recipe adaptation scope

Matched-recipe adaptation factorials require source prediction caches and aggregate summary JSONs that are not included in the supplementary artifact. We therefore do not tabulate those values and do not use them to support the main claims. The released adaptation evidence is limited to explicit released traces under `per_case_dice_adapted` and the aggregate summaries named in the surrounding appendix sections.

## A26 Item-difficulty residualisation and held-out nulls

The main-text Section 5 reports a cross-fitted residualised $\Delta$ diagnostic on all five primary rows. Three distinct residualisation procedures appear in this appendix and we tabulate them in order of provenance: (i) the *naive* all-bundle null, included only as a leakage warning; (ii) the *random A-vs-B family-split holdout* on Cup and ACDC (the existing `r7_item_difficulty_holdout.json` diagnostic, Table 26 below); and (iii) the *cross-fitted broad-family-leaveout* on all five primary rows (Table 25, the headline 5-row diagnostic referenced in the main text). The cross-fitted procedure is the principled all-row diagnostic and supplies the main text's $\Delta$=0.830–0.976 residualised range; the held-out-partition table is retained as the historical Cup/ACDC-only diagnostic from `r7_item_difficulty_holdout.json`.

**Cross-fitted broad-family-leaveout (headline, all five rows).** For each trace, item difficulty is estimated from traces outside that trace's broad upstream family, the trace is residualised by OLS against this leave-out estimator, and the Table 1 same/cross rule is recomputed on residuals. The residualised gap is positive on all five primary rows. Residualised gaps *exceed* raw gaps because the cross-fitted regressor removes shared item-difficulty signal that both same- and cross-bundle pairs already share at the per-case level; same-bundle residual continues to capture bundle-specific systematic deviations not absorbed by case difficulty, while cross-bundle residual decorrelates faster. The attenuation ratio (residual gap / raw gap) is $1.37$–$3.68$ across the five tasks.

Table 25: **Cross-fitted item-difficulty residualisation, all five primary rows.** Each trace's item-difficulty regressor is estimated using only traces from broad upstream families *other than* the trace's own family; the same-vs-cross Pearson gap is then recomputed on residuals. Residualised gaps exceed raw gaps because the cross-fitted regressor removes shared item-difficulty signal that both same- and cross-bundle pairs already share at the per-case level; same-bundle residual continues to capture bundle-specific systematic deviations not absorbed by case difficulty, while cross-bundle residual decorrelates faster. Source: `results/_merged/diagnostics/item_difficulty_robustness.json` (schema `fmpool_item_difficulty_robustness_v1`); $B$=2,000 case bootstrap.

| Task | Raw $\Delta$ 95% CI | Residualised $\Delta$ 95% CI | Attenuation ratio | Functional pool |
|---|---|---|---|---|
| Kvasir | [0.211, 0.320] | [0.895, 1.008] | 3.68 | 11-bundle |
| ACDC LV | [0.289, 0.355] | [0.950, 1.000] | 3.05 | 10-bundle |
| BraTS foreground | [0.302, 0.329] | [0.910, 0.932] | 2.92 | 10-bundle |
| RIGA Cup | [0.395, 0.556] | [0.782, 0.882] | 1.75 | 9-bundle |
| RIGA Disc | [0.569, 0.687] | [0.805, 0.920] | 1.37 | 11-bundle |

**Naive (leaked) null.** Define difficulty $d_i = 1 - \frac{1}{K}\sum_{k=1}^{K} D_k^{(42)}(i)$ where $D_k^{(42)}$ is bundle $k$'s per-case Dice at seed 42, then residualise each $D_k^{(s)}$ against $d$ by OLS and then evaluate $\bar{r}$. This instantiates one item-difficulty null inspired by Jo et al. [2026], but has a structural flaw: each bundle's Dice is a component of its own regressor (since $d$ averages over all 8 bundles including $k$), so residuals are mechanically constrained to sum to approximately zero across bundles at each case. For $K$ bundles at seed 42, cross-bundle residual correlations are pushed toward $-1/(K-1) \approx -0.14$, and we observe cross-family residuals of $-0.088$ (Cup) and $-0.104$ (ACDC), which sharply widens $\Delta$ compared to the raw statistic. This is a mechanical leakage artifact, not a cleaner measurement.

**Random A-vs-B family-split holdout (two-task null).** To remove the leakage we split the 8 bundles into halves $A$ and $B$, estimate $d$ from $A$'s seed-42 Dice only, and residualise $B$'s Dice on this $d$. The $\bar{r}$ statistics and $\Delta$ are then computed on $B$. The released diagnostic JSON records a fixed set of 10 held-out partitions and reports the mean and range of the resulting family-grouped $\Delta$ values across those partitions. This procedure is procedurally distinct from the cross-fitted broad-family-leaveout above: it does a random within-task split of the panel rather than a leave-one-broad-family-out cross-fit, and it covers only Cup and ACDC.

Table 26: **Two-task item-difficulty residualisation diagnostics (random A-vs-B family-split holdout).** The naive null is included only as a leakage warning. Raw values use this diagnostic's family grouping and are not Table 1 values; the held-out-partition null avoids leakage on Cup/ACDC only. This is procedurally distinct from the cross-fitted broad-family-leaveout in Table 25. Source: `r7_item_difficulty_holdout.json`.

| Null | RIGA Cup $\Delta$ | ACDC $\Delta$ | Notes |
|---|---|---|---|
| Raw | 0.499 | 0.600 | Family grouping |
| Naive difficulty (all-bundle leaked) | 0.886 | 0.772 | Cross-family residuals$\approx -0.10$ (leakage signature) |
| **Held-out difficulty (released held-out partitions)** | **0.639** | **0.622** | Family grouping; $n$=10 per task |
| Held-out range | [0.482, 0.823] | [0.472, 0.941] | Min / max across partitions |

The two-task held-out estimate is above raw $\Delta$ on Cup ($+0.14$) and remains slightly above raw on ACDC ($+0.02$). Under this *one-directional* null, item difficulty does not explain the gap away. The naive inflation was a mechanical leakage artifact regardless.

**Residualisation caveat.** The released held-out partitions are one-directional diagnostics rather than a symmetric enumeration over every possible difficulty/evaluation split. They should be read as evidence that this particular leakage-free residualisation does not absorb the gap, not as a two-sided causal decomposition of difficulty, architecture, and pretraining. The all-row cross-fitted diagnostic broadens coverage to every primary row, but it still estimates difficulty from the same benchmark panel and therefore remains a robustness check rather than a pre-registered causal adjustment.

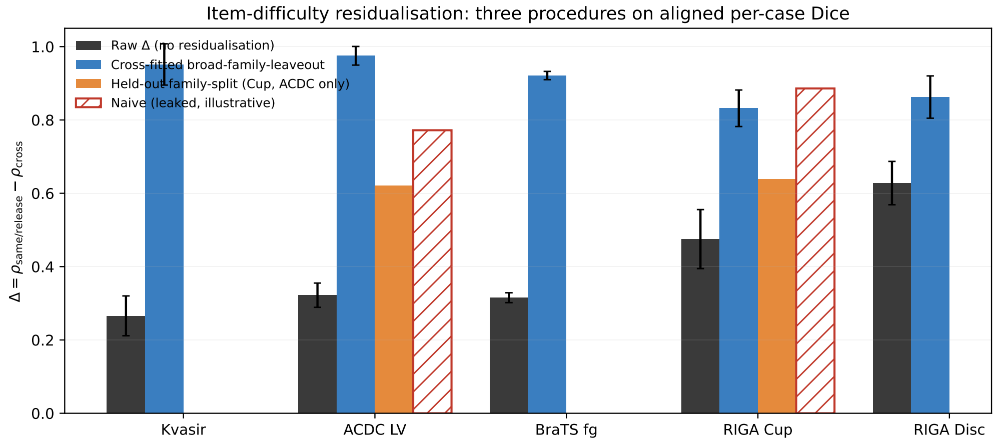


Figure 3: **Three item-difficulty residualisation procedures on aligned per-case Dice.** Each procedure operates on the same released traces but uses a different difficulty regressor. *Raw* (dark grey) is Pearson $\Delta$ without residualisation, recomputed per task at the audit's family grouping. *Cross-fitted broad-family-leaveout* (blue, all five primary rows) estimates each trace's difficulty regressor from traces outside the trace's own broad upstream family; the residualised gap CIs are taken from `item_difficulty_robustness.json` (`cross_fitted_item_difficulty_floor_0.30.bootstrap`). *Held-out family-split* (orange, Cup and ACDC only) is the random A-vs-B split-holdout from `r7_item_difficulty_holdout.json`; mean across released partitions. *Naive (leaked, illustrative)* (red hatched) is the all-bundle null whose mechanical leakage is documented in the prose above. The cross-fitted bars are the procedure that supplies the main-text $\Delta$=0.830–0.976 residualised range; the held-out family-split bars are a procedurally distinct two-task companion. Error bars are 95% case-bootstrap intervals where reported. Source script: `code/scripts/build_diffleak_procedures_figure.py`.

## A27 Correlation-equivalent effective pool size $M_{\text{eff}}$

**Definition.** For a pool of $M$ members whose per-case failure indicators $F_1, \ldots, F_M$ at threshold $\tau$ are exchangeable with mean pairwise Pearson correlation $\rho_{\text{fail}}$, the variance of the per-case mean indicator is $\sigma_F^2[1 + (M-1)\rho_{\text{fail}}]/M$ where $\sigma_F^2$ is the marginal variance. Equating this with the variance of a hypothetical independent pool of size $M_{\text{eff}}$, i.e. $\sigma_F^2/M_{\text{eff}}$, gives

$$M_{\text{eff}} \;=\; \frac{M}{1 + (M-1)\,\rho_{\text{fail}}}.$$

This is the equicorrelation approximation also used in climate-model ensembles ("effective number of models"). We report it as a second-moment readout of the audit; it is *not* a literal independent-member count. The reported $M_{\text{eff}}$ values follow algebraically from the primary correlation values in Table 1; we report them as a compact summary of the primary same-vs-cross gap, not as additional independent evidence.

**Reading the table.** The scorer uses the same estimand as the text: a diverse pool is four distinct FMs with one decoder-seed member from each FM. Under that definition, the nominal $M$=4 diverse pools yield $M_{\text{eff}} \approx$ 2–3 under the mean equicorrelation readout, not four. The $\rho_{\max}$ column shows that the most redundant valid diverse tuples have lower conservative readouts (1.30–1.69), while same-FM seed pools are near one under the same calculation. The exact level varies by task, tuple, and failure threshold, so we use $M_{\text{eff}}$ as a dependence readout, not as a literal deployment objective.

**Pairwise $\rho_{\text{fail}}$ heterogeneity disclosure (equicorrelation caveat).** The $M_{\text{eff}}$ formula assumes constant pairwise correlation across member pairs (equicorrelated pool). In practice the per-pair $\rho_{\text{fail}}$ varies across the bundles in each task. From the `m_eff/*.json` outputs, mono-pool $\rho_{\text{fail}}$ across the per-task functional bundles has mean / SD / range: Kvasir mean 0.865, SD 0.103, range $[0.601, 0.948]$ ($n$=11); ACDC LV mean 0.847, SD 0.122, range $[0.518, 0.931]$ ($n$=10); BraTS foreground mean 0.860, SD 0.117, range $[0.553, 0.958]$ ($n$=10); RIGA Cup mean 0.833 over 7

Table 27: **Threshold-specific effective pool-size diagnostic at $\tau$=0.85.** $M_{\text{eff}}$ uses an equicorrelation approximation for binary failure indicators and is not a literal independent-member count. Same-bundle pools are each FM's completed decoder seeds. Diverse rows enumerate every 4-bundle composition of the filtered functional pool and every available one-seed-per-FM tuple. $\rho_{\max}$ is the largest valid diverse-pool $\rho_{\text{fail}}$ over that released enumeration and gives a conservative lower $M_{\text{eff}}$ readout for the most redundant valid tuple. Undefined zero-variance failure-correlation tuples are excluded rather than imputed. Source: `results/_merged/m_eff/*.json`.

| Task | Func. bundles | Mono mean[†] | Diverse $\bar{\rho}_{\text{fail}}$ | $\rho_{\max}$ | Mean $M_{\text{eff}}$ | $M_{\text{eff}}(\rho_{\max})$ | Valid tuples |
|---|---|---|---|---|---|---|---|
| Kvasir | 11 | 1.12 | 0.333±0.104 | 0.625 | 2.05±0.32 | 1.39 | 84,480 / 84,480 |
| ACDC LV | 10 | 1.14 | 0.298±0.075 | 0.528 | 2.14±0.24 | 1.55 | 53,760 / 53,760 |
| BraTS foreground | 10 | 1.13 | 0.273±0.089 | 0.506 | 2.24±0.32 | 1.59 | 53,760 / 53,760 |
| RIGA Cup | 9 | 1.16 | 0.246±0.121 | 0.695 | 2.43±0.63 | 1.30 | 31,488 / 32,256 |
| RIGA Disc | 11 | 1.17 | 0.126±0.067 | 0.455 | 2.97±0.44 | 1.69 | 84,480 / 84,480 |

[†]Mean across per-FM seed pools after excluding undefined zero-variance failure indicators. The RIGA Cup row has 7/9 defined mono pools and the RIGA Disc row has 10/11; the other rows have all mono pools defined.

defined mono pools, and RIGA Disc mean 0.812 over 10 defined mono pools. Undefined rows occur when an FM's $\tau$=0.85 failure indicator is constant, giving zero-variance Pearson. Diverse 4-of-$K$ compositions span a wider $\rho_{\text{fail}}$ distribution, which is why Table 27 reports both means and the released-enumeration $\rho_{\max}$ rather than selecting a single representative pool.

## A28 Shared-backbone random-effects representation

**Model.** For each FM $m$ in a frozen-feature pool and test case $i$, let the per-case score (or its logit) decompose as

$$s_m(i) \;=\; u(i) \;+\; a_{\text{arch}(m)}(i) \;+\; p_{\text{family}(m)}(i) \;+\; \varepsilon_m(i),$$

where $u$ is task-inherent item difficulty, $a$ is a zero-mean architecture-family random effect, $p$ is a zero-mean pretraining-family random effect nested within architecture, and $\varepsilon_m$ is an FM-specific adaptation noise. Assume $u, a, p, \varepsilon$ are mutually uncorrelated with variances $\sigma_u^2, \sigma_a^2, \sigma_p^2, \sigma_\varepsilon^2$. Then the pairwise covariance of $s_m, s_{m'}$ takes a nested form:

$$\mathrm{Cov}(s_m, s_{m'}) \;=\; \begin{cases} \sigma_u^2 + \sigma_a^2 + \sigma_p^2, & \text{(same family)} \\ \sigma_u^2 + \sigma_a^2, & \text{(same architecture, different pretraining)} \\ \sigma_u^2, & \text{(different architecture).} \end{cases}$$

**Diagnostic expectation.** Under this second-moment description, and absent parameter cancellations, one expects the ordering

$$\rho_{\text{within-FM}} \;\geq\; \rho_{\text{same-arch-diff-pretraining}} \;\geq\; \rho_{\text{cross-arch}}$$

as a qualitative diagnostic rather than a theorem. Empirically on 5 same-architecture-family ViT variants (Table 3) we observe the ordering on Cup, and on ACDC after applying the functional filter: $0.988 \geq 0.807 \geq 0.639$; the all-5 ACDC row is a sensitivity to including nonfunctional CLIP-B. This is consistent with the predicted ordering and with trajectory/same-basin evidence from deep ensembles and model soups [Fort et al., 2019, Neyshabur et al., 2020, Wortsman et al., 2022]; Abe et al. [2024] is better read here as motivation that predictive diversity can remain constrained in high-capacity ensembles, not as a trajectory-bound theorem.

**Implication for $M_{\text{eff}}$.** When the binary failure thresholds preserve the same broad ordering as the continuous Dice scores, $M_{\text{eff}}$ is monotone decreasing in the observed $\rho_{\text{fail}}$ by definition. The empirical diagnostics therefore suggest the descriptive ordering $M_{\text{eff}}^{\text{within-FM}} \leq M_{\text{eff}}^{\text{same-arch}} \leq M_{\text{eff}}^{\text{cross-arch}}$ in the tested regime, but the paper estimates this quantity from released per-case traces rather than inferring it from labels alone. Same-bundle seed pools are same-FM by construction and sit near the lower end of the measured $M_{\text{eff}}$ range; diverse pools that cross both architecture and pretraining bundles sit closer to the upper end.

**Caveats.** This is a second-moment model at the per-case-score level, not a first-principles claim about SGD dynamics. The random-effects representation is descriptive; it gives a language for the observed ordering and a hook for bounding pool redundancy, not a proof that all shared-pipeline pools must have the same structure. The item-difficulty residualisation of Appendix A26 removes $u$ and leaves $a + p + \varepsilon$; our held-out null is consistent with $a + p$ contributing after $u$ is removed.

## A29 Architecture-confound diagnostics

The main-text Table 3 reports the released 5-variant same-architecture-family ViT / different-pretraining diagnostic on Cup and ACDC. A secondary Kvasir-SEG replication used the same 5 ViT-B variants, but its aggregate JSON is outside the released artifact, so it is not tabulated as released evidence here.

Table 28: **Appendix detail for the same-architecture ViT diagnostic.** Within-FM is the decoder-seed floor; same-architecture-family ViT cross-pretraining holds the broad architecture family fixed while varying pretraining. This expands Table 3 by showing the all-5 ACDC sensitivity and the functional-filtered readout. Source: released aggregate diagnostic `results/_merged/diagnostics/r7_arch_confound_analysis.json`; raw NPZ prediction caches are outside the released artifact.

| Task | ViT subset | Used | Seed floor | ViT cross-pretrain | Primary cross | Gap |
|---|---|---|---|---|---|---|
| RIGA Cup | all 5 ViTs | 5/5 | 0.968 | 0.680 | 0.361 | 0.288 |
| ACDC LV | all 5 ViTs | 5/5 | 0.965 | 0.631 | 0.639 | 0.334 |
| ACDC LV † | CLIP-B dropped | 4/5 | 0.988 | 0.807 | 0.639 | **0.182** |

The same-architecture cross-pretraining separation is 0.18–0.29 across the released Cup/ACDC diagnostics under the paper's functional filter († ACDC row drops the non-functional CLIP-B at Dice$=0.21<0.30$, which otherwise compresses same-arch cross $\bar{r}$ and inflates the separation to 0.33). The direction is preserved, but the claim is mechanism-diagnostic rather than causal. Results are consistent with the random-effects representation in Appendix A28.

## A30 Same-family cross-checkpoint decomposition

The main text explicitly cautions that the primary "same" column is dominated by same-bundle different-decoder-seed pairs. To test whether the signal collapses to that seed floor, we recompute the released same-task per-case traces, including additional same-family checkpoint variants beyond the primary pool, under a three-stratum decomposition: (i) same-FM decoder-seed pairs, (ii) distinct checkpoints within a broad upstream family (DINOv2-B/L/S, OpenAI CLIP B/16–B/32–L/14, ConvNeXt-T/S/B, and ResNet-18/34/50/101 where functional), and (iii) cross-family pairs. This is a diagnostic broad-family grouping rather than a causal architecture/pretraining decomposition. The distinct-checkpoint same-family gap remains positive on all five primary rows, but it is substantially below the same-bundle seed floor.

Table 29: **Same-family cross-checkpoint diagnostic.** Same-FM seed pairs form the upper dependence floor; distinct-checkpoint same-family pairs remain above the diagnostic cross-family stratum but below same-bundle seeds. The `Diagnostic cross` column is not the Table 1 cross column. Source: `results/_merged/diagnostics/cross_checkpoint_family.json`.

| Task | Same-bundle | Same-family checkpoint | Diagnostic cross | Checkpoint gap | 95% CI |
|---|---|---|---|---|---|
| Kvasir | 0.987 | 0.777 | 0.697 | 0.080 | [0.043, 0.122] |
| ACDC LV | 0.986 | 0.750 | 0.616 | 0.134 | [0.113, 0.155] |
| BraTS foreground | 0.977 | 0.682 | 0.589 | 0.093 | [0.085, 0.100] |
| RIGA Cup | 0.963 | 0.537 | 0.271 | 0.267 | [0.204, 0.332] |
| RIGA Disc | 0.971 | 0.421 | 0.194 | 0.227 | [0.149, 0.293] |

CIs are case-bootstrap intervals with $B$=2,000. The cross column is the diagnostic broad-family cross stratum from this expanded checkpoint grid, not the Table 1 analytic-pool cross-family column. For the slice-derived rows, subject-aggregated checkpoint-family gaps remain positive: ACDC 0.166 [0.091, 0.262] and BraTS 0.098 [0.078, 0.119]. BiomedCLIP is kept separate from OpenAI CLIP because it changes both pretraining corpus and objective. The OpenAI CLIP same-family row mixes CLIP-B/16 QuickGELU-style checkpoints with the released CLIP-L/14 `open_clip` standard-GELU dense wrapper, so that family-specific readout also absorbs this loader/activation difference.

## A31 Additional mechanism and scope diagnostics

This section reports seven additional scope analyses that close axes around the main result: decoder capacity, random-initialized CNNs, loss/augmentation, fold-reslicing, BraTS train-size, BraTS multimodal input, and MC dropout. The released artifact contains the corresponding per-case Dice traces and the CPU-only summary scripts used for these appendix tables. All numbers are computed directly from the per-case traces; per-FM means use seed averaging where multiple seeds are present, and Pearson $\bar{r}$ is symmetric multi-seed.

### A31.1 Decoder-capacity sweep

We rerun the Section 6 (C1) decoder sweep at four explicit head designs spanning $769$ to $3.94$M trainable parameters: `linear_1x1`, `mlp_2layer`, `unet_lite`, and `transformer_decoder`. Two FMs (DINOv2-B, BiomedCLIP) $\times$ two seeds $\{42, 43\}$ on RIGA Cup. Across the $\sim 5{,}000\times$ capacity range, pool-mean Dice ranges over $0.725$–$0.848$ but cross-decoder within-FM Pearson $\bar{r}$ remains $0.69$ on average ($n$=48 pairs across the four head designs and the two FMs)—above the RIGA Cup functional-pool cross-family reference of $0.361$ (Table 1). Head capacity changes accuracy and some correlations, but does not eliminate same-backbone co-failure in this diagnostic.

Table 30: **Head-capacity sweep on RIGA Cup** (2 FMs $\times$ 2 seeds, 30-epoch frozen probe). Pool-mean Dice and trainable-parameter count per head design; the diagnostic's within-FM cross-head correlation summary is $\bar{r}$=0.690 (range 0.510–0.896), above the Table 1 RIGA Cup cross reference of 0.361.

| Head design | Pool-mean Dice | Trainable params |
|---|---|---|
| `linear_1x1` | 0.725 | 769 |
| `mlp_2layer` | 0.810 | 197,121 |
| `unet_lite` | 0.848 | 3,319,105 |
| `transformer_decoder` | 0.805 | 3,940,865 |

### A31.2 Random-init CNN decorrelation

To check whether within-vs-cross structure persists in the absence of pretraining, we rerun the released random-init diagnostic on two CNN backbones (ConvNeXt-Tiny, ResNet-50) trained from scratch, 8 seeds $\{42-49\}$, on ACDC LV and RIGA Cup. Even with no shared pretraining, within-FM (cross-seed) $\bar{r}$ stays clearly above cross-FM (different-architecture) $\bar{r}$: $0.916$ vs. $0.830$ on ACDC LV ($\Delta$=0.086), $0.841$ vs. $0.720$ on RIGA Cup ($\Delta$=0.121). The gap is much smaller than the frozen-FM Table 1 numbers ($0.834/0.361$ on Cup, $\Delta$=0.473), but it is still positive, consistent with architecture and shared recipe contributing to within-vs-cross structure in this diagnostic.

Table 31: **Random-init CNN same/cross diagnostic.** Two CNN backbones $\times$ eight seeds; 30 epochs from scratch; same $1\times1$ probe protocol. Positive but smaller gaps suggest architecture/recipe dependence even without shared pretrained weights. Diagnostic only.

| Task | within-FM $\bar{r}$ | cross-FM $\bar{r}$ | $\Delta$ |
|---|---|---|---|
| ACDC LV | 0.916 | 0.830 | 0.086 |
| RIGA Cup | 0.841 | 0.720 | 0.121 |

### A31.3 Loss/augmentation sensitivity

The Section 4 protocol uses BCE loss, $224$ inputs, no augmentation. To check whether the gap survives a closer-to-standard frozen-head loss/augmentation recipe, we evaluate the seven-bundle subset {DINOv2-B/S, CLIP-L, ConvNeXt-T, EfficientNet-B0, ResNet-50/18} $\times$ 4 seeds with combined soft Dice+BCE loss and paired {hflip, rot$\pm15°$} augmentation. All runs still keep the encoder frozen, use the same $1\times1$-conv decoder, Adam $10^{-3}$, and encoder-native $224\times224$ input; this is not a full clinical segmentation pipeline. The original RIGA Cup row remains the primary loss/augmentation diagnostic in Appendix A18; we also report three task-level diagnostic extensions (RIGA Disc, ACDC LV, and an ISIC-2018 Task 1 fixed 400-image held-out subset) to check whether the positive gap is an isolated Cup result. ISIC here is a supplementary per-case Dice diagnostic on a fixed 400-image subset and is not the official ISIC challenge test/server protocol or threshold-Jaccard leaderboard score.

Across all four loss/augmentation rows, the Dice+BCE+augmentation recipe leaves a positive within-vs-cross gap ($\Delta$ = 0.326–0.617). The magnitude varies by task: ACDC remains lower

Table 32: **Loss/augmentation sensitivity on RIGA Cup, 7 FMs $\times$ 4 seeds.** This is a closer-to-standard frozen-head recipe using Dice+BCE and simple augmentation, not a full clinical segmentation pipeline; all rows remain frozen $1\times1$ heads at $224\times224$. Per-FM seed-averaged Dice; pool $\bar{r}$ at the bottom. CLIP-L uses the same caveated OpenCLIP `ViT-L-14/openai` dense-token wrapper as the primary bundle.

| FM | mean Dice | $\sigma$ (seed) |
|---|---|---|
| DINOv2-B | 0.758 | 0.005 |
| DINOv2-S | 0.733 | 0.003 |
| CLIP-L | 0.720 | 0.006 |
| EfficientNet-B0 | 0.626 | 0.002 |
| ResNet-50 | 0.618 | 0.010 |
| ConvNeXt-T | 0.583 | 0.004 |
| ResNet-18 | 0.526 | 0.027 |
| Pool: within $\bar{r}$=0.903, cross $\bar{r}$=0.556, $\Delta$=0.347 [0.275, 0.428]. | | |

Table 33: **Task-level loss/augmentation extensions.** Same seven-bundle, four-seed frozen-head Dice+BCE+augmentation recipe as Table 32. These rows are not primary rows. Dice columns summarise per-bundle seed-averaged Dice across the seven bundles. The ISIC row is a fixed 400-image-subset held-out diagnostic, not the official ISIC test/server evaluation.

| Task | $N_{\text{test}}$ | FM-mean Dice | FM min | FM max | $\Delta$ | 95% CI |
|---|---|---|---|---|---|---|
| RIGA Cup | 95 | 0.652 | 0.526 | 0.758 | 0.347 | $[0.275, 0.428]$ |
| RIGA Disc | 95 | 0.853 | 0.756 | 0.922 | 0.617 | $[0.535, 0.693]$ |
| ACDC LV | 361 | 0.565 | 0.389 | 0.740 | 0.326 | $[0.291, 0.366]$ |
| ISIC-2018 staged | 80 | 0.864 | 0.801 | 0.918 | 0.473 | $[0.375, 0.562]$ |

accuracy and lower-gap than the RIGA/ISIC rows, while RIGA Disc shows the largest extension gap. These rows are treated as scope checks rather than primary evidence because they use a seven-FM subset and, for ISIC, a fixed subset/split; they indicate that the simplified BCE/no-augmentation protocol is not the sole source of the co-failure signal.

### A31.4 Five-fold CV robustness

The fold-reslicing analysis is reported with the nnU-Net complements in Appendix A17 and Table 17. The key scope point is that the released artifact has one decoder seed per fold; it supports split-stability of cross-bundle $\bar{r}$, not a within-fold vs cross-fold same-backbone estimator.

### A31.5 BraTS foreground train-size sweep

This train-size diagnostic covers the five BraTS foreground bundles with completed feature caches (DINOv2-B, BiomedCLIP, CLIP-L/14, EfficientNet-B0, ResNet-50). It uses one decoder seed (42) and fractions $\{0.10, 0.25, 0.50, 1.00\}$ of the same cached training set, so it is an *accuracy/saturation* check rather than a within-vs-cross dependence estimator. The result is non-monotone: the 5-bundle mean Dice rises from $0.579$ at 10% to $0.621$–$0.622$ at 25–50%, then stays flat/slightly lower at full data ($0.617$). Thus, in this cached five-bundle subset, adding more 2D FLAIR slices improves the weakest low-data setting but does not by itself create a new high-performing, decorrelated pool; the correlation claim still has to be audited directly rather than inferred from sample size.

Table 34: **BraTS train-size accuracy/saturation diagnostic, one seed only** (5 cached bundles, seed 42). Mean/min/max Dice are across bundles at each train fraction; this table does not estimate within/cross dependence because each bundle has one seed.

| Train fraction | Cells | Mean Dice | Min Dice | Max Dice |
|---|---|---|---|---|
| 0.10 | 5 | 0.579 | 0.418 | 0.721 |
| 0.25 | 5 | 0.621 | 0.455 | 0.750 |
| 0.50 | 5 | 0.622 | 0.436 | 0.751 |
| 1.00 | 5 | 0.617 | 0.458 | 0.737 |

### A31.6 BraTS multimodal RGB derivative

This analysis checks whether the BraTS foreground co-failure signal is an artifact of using a unimodal FLAIR derivative. We build a deliberately simple three-channel 2D derivative with the same held-out subjects and top-10 foreground-area slice rule as Table 1, but pack three MRI modalities into RGB channels: R=T1c, G=T2w, B=T2-FLAIR. This is *not* the official 3D BraTS challenge protocol; it is a controlled frozen-feature stress test of the input-channel choice.

The evaluated grid contains 20 cells (5 FMs $\times$ 4 seeds). Multimodal input narrows the dependence gap relative to the unimodal Table 1 BraTS row, but does not remove it: case-level within $\bar{r}=0.904$, cross $\bar{r}=0.654$, $\Delta=0.249$ [0.238, 0.262]. Aggregating each subject's slices before correlation gives within $\bar{r}=0.946$, cross $\bar{r}=0.772$, $\Delta=0.174$. The conservative interpretation is therefore narrower than "BraTS input modality does not matter": adding T1c/T2w raises cross-family agreement and shrinks the gap, yet same-backbone/family dependence remains positive under the same frozen-head audit.

Table 35: **Simple three-channel BraTS 2D derivative** (R=T1c, G=T2w, B=T2-FLAIR; 5 FMs $\times$ 4 seeds). This is the same 2D frozen-head audit, not the official 3D BraTS multimodal challenge protocol. Dice is seed-averaged by FM; dependence uses the same Pearson estimator as Table 1.

| FM | Mean Dice | Min Dice | Max Dice | Seeds |
|---|---|---|---|---|
| BiomedCLIP | 0.713 | 0.669 | 0.739 | 4 |
| DINOv2-B | 0.703 | 0.675 | 0.721 | 4 |
| DINOv2-S | 0.702 | 0.690 | 0.714 | 4 |
| CLIP-L/14 | 0.686 | 0.682 | 0.692 | 4 |
| ResNet-50 | 0.461 | 0.439 | 0.473 | 4 |

Case-level pool: within $\bar{r}=0.904$, cross $\bar{r}=0.654$, $\Delta=0.249$ [0.238, 0.262].
Subject-aggregated pool: within $\bar{r}=0.946$, cross $\bar{r}=0.772$, $\Delta=0.174$.

### A31.7 MC dropout as a within-family decorrelator

A natural pool-design question is whether MC dropout [Gal and Ghahramani, 2016] can act as a lightweight within-family decorrelation lever. We rerun DINOv2-B on RIGA Cup with three dropout rates $p \in \{0.1, 0.2, 0.4\}$, 4 seeds, 10 MC samples per case. The deterministic within-FM cross-seed $\bar{r}\approx$0.97–0.99; under MC averaging, $\bar{r}$ drops to $0.95$ ($p$=0.1), $0.93$ ($p$=0.2), $0.89$ ($p$=0.4). The largest tested drop is $\approx 0.10$ and remains materially smaller than the same-architecture-family ViT-vs-primary-cross separation seen on Cup (Table 3). MC dropout is therefore a weak within-family lever in this diagnostic; consistent with Appendix A13, it does not approach the architecture-family separation.

**Higher-$p$ and any-draw scope.** A separate, non-released MC-dropout probe at higher dropout rates suggested lower any-draw correlations than the averaged readout above, but its raw traces are outside the released artifact and its scorer differs from the deterministic-vs-MC-averaged readout used in this section. We therefore do not tabulate those values; the released MC-dropout evidence is the table above, which uses the bundled per-case traces.

Table 36: **MC-dropout diagnostic.** DINOv2-B/RIGA Cup with four seeds and 10 MC samples per case; correlations are computed on MC-averaged per-case Dice. MC dropout provides modest decorrelation but remains far above the Table 1 RIGA Cup cross-family reference of 0.361. Diagnostic only.

| Dropout $p$ | MC mean Dice | Det. within $\bar{r}$ | MC within $\bar{r}$ | $H_{\text{pred}}$ |
|---|---|---|---|---|
| 0.1 | 0.666 | 0.972 | 0.947 | 0.062 |
| 0.2 | 0.657 | 0.972 | 0.925 | 0.067 |
| 0.4 | 0.633 | 0.987 | 0.893 | 0.076 |

## A32 Release manifest and artifact metadata

The artifact bundle includes the primary-task result JSONs, source code, metadata, selected figure / appendix inputs, and selected aggregate diagnostic JSONs needed to audit the numerical claims from saved artifacts where redistribution is permitted. The primary rows are stored under `results/_merged`: the per-seed `per_case_dice` JSONs carry `test_ids`, `per_case_dice`, FM name, seed, and checksum fields; `within_cross/*.json`, `m_eff/*.json`, and `paper_table.json` store the primary summaries. The same directory also includes `subject_level/*.json` for the ACDC/BraTS cluster checks, `calibration_split/*.json` as split-half sensitivity (not the primary protocol), `provenance_summary.json` for source counts and case-unit metadata, and `diagnostics/*.json` for the pixel-error, matched-lift, architecture, quality-controlled pair-regression, item-difficulty, and held-out aggregate diagnostics used only as secondary evidence. The bundle does *not* include raw medical images, raw labels, full prediction NPZs, feature caches, or held-out raw prediction arrays; full split manifests beyond the case IDs embedded in the result JSONs are not included in this artifact release.

**ID and source-label scope.** Kvasir rows expose only image-level file identifiers in the released `test_ids`; no patient, video, or near-duplicate grouping identifier is available in the released trace bundle, so we do not claim patient-level duplicate control for that row. RIGA Cup/Disc rows retain deterministic hashed alignment identifiers and non-identifying source/task labels only to make the $n$=95 Magrabia-source test split auditable. Raw upstream folder-name components are not needed for this analysis and are not redistributed as case identifiers; the released identifiers are not validated demographic annotations and must not be used for subgroup, gender, or individual-level analysis.

**Encoder-loader boundary.** The released traces should be read as frozen-feature probes through the bundled loader wrappers, not as official vendor inference runs. In particular, the `clip_vitl14` traces use the version-pinned `ViT-L-14/openai` path exposed by the installed `open_clip` package; `open_clip` also exposes a `ViT-L-14-quickgelu/openai` model definition and warns that OpenAI weights are associated with QuickGELU. We therefore treat CLIP-L/14 as the bundle's documented dense-feature wrapper and do not use it to make a claim about official OpenAI CLIP inference accuracy. The code keeps this loader unchanged so that the released per-case traces remain reproducible.

**Compute and environment scope.** The artifact-verification path is CPU-only: the bundled verification script and the individual `compute_*` scripts read released JSON traces and finish in minutes on a typical workstation. Raw-image reproduction is heavier and requires separate upstream data/checkpoint access. The reported training runs used a shared GPU cluster with NVIDIA A100-80GB, A6000, and L40S devices; feature extraction time depends on backbone and task size, and cached frozen-head training is lightweight relative to feature extraction. We therefore document resource classes and released trace-level commands rather than claiming a single exact GPU-hour total because raw-image retraining is outside the submitted artifact's reproduction scope.

Table 37: **Compute scope for reproducibility.** The submission supports trace-level recomputation directly; raw-image training resources are provided as reproduction guidance because upstream data, checkpoints, and feature caches are not redistributed.

| Reproduction target | Required compute | Notes |
|---|---|---|
| Primary and diagnostic summary JSONs | CPU, minutes-scale | Run `PYTHONPATH=code/src python3 code/scripts/verify_artifact.py`; no raw images, checkpoints, Torch, or GPU required. |
| Frozen-feature primary traces from raw data | A100-80GB/A6000/L40S-class GPU plus storage for upstream data and features | Feature extraction is the main cost; cached $1\times1$ decoder-head training uses Adam for 30 epochs and is comparatively lightweight. |
| Appendix architecture, adaptation, and nnU-Net complements | Same cluster GPU classes; optional MONAI/nnU-Net environments where applicable | These are diagnostic or limited-scope runs. The bundled trace-level artifact is sufficient to recompute reported released summaries without rerunning them. |

**Functional-floor members are retained in the source files.** The paper applies a Dice$\geq$0.30 functional floor downstream in the estimator, not at result ingestion. Thus the merged `per_case_dice` tree can list more completed JSONs than the analytic pool used by Table 1; the extra cross-checkpoint members support Appendix A30. For example, ACDC LV has 68 merged source JSONs

Table 38: **Trace-level claim-to-artifact map.** The artifact supports recomputation of primary audit statistics from derived per-case Dice traces. It does not reproduce raw-image training unless users separately obtain upstream datasets and checkpoints. Aggregate-only diagnostics are marked as such rather than promoted to primary evidence; exact commands and checksums are in `README.md`, `metadata/MANIFEST.md`, and `metadata/SHA256SUMS`.

| Claim/readout | Released source | Recompute / limitation |
|---|---|---|
| Primary same-vs-cross gaps, CIs, and filtered functional pools | `paper_table.json`, `within_cross/*.json`, `per_case_dice/*.json` | `PYTHONPATH=code/src python3 code/scripts/verify_artifact.py`; case IDs and per-case Dice are included, raw images and masks are not; trace-level recomputation only. |
| Subject, floor, ICC, permutation, event, and quality-control robustness | `subject_level/*.json`, `audit_sensitivity.json`, `failure_event_summary.json`, `quality_controlled_pairs.json` | The verification script calls `compute_all.py`, `compute_audit_sensitivity.py`, `compute_failure_event_summary.py`, `compute_quality_controlled_pairs.py`; these are estimator checks, not new training runs. |
| Item-difficulty and cross-checkpoint diagnostics | `item_difficulty_robustness.json`, `cross_checkpoint_family.json`, `r7_arch_confound_analysis.json` | `compute_item_difficulty_robustness.py` and `compute_cross_checkpoint_family.py`; diagnostic evidence for alternative explanations, not causal identification. |
| Pixel/lift, same-backbone, held-out, and nnU-Net scope rows | `diagnostics/pixel_error/*.json`, `diagnostics/matched_lift/*.json`, appendix trace files, `nnunet/*.json`, `nnunet_case_riga_2d_100ep/*.json` | Released summaries plus the RIGA case-identical nnU-Net per-model traces; other full-pipeline comparators remain aggregate or task-level only. |

Table 39: **Trace inventory and analytic-pool provenance.** `Merged` counts all completed (bundle, decoder-seed) JSON entries under `results/_merged/per_case_dice`; Table 1 uses the original 44-trace primary source subset and applies the functional-floor pool shown in the final column. Same/cross pair counts for the analytic pool are reported in Table 1. The release key `brats_wt` denotes the 2D BraTS `seg>0` foreground derivative used in this paper; it is not the official 3D BraTS WT challenge target.

| Task | $N$ | Merged bundles | Merged members | Table 1 source | Analytic pool |
|---|---|---|---|---|---|
| `kvasir` | 120 | 17 | 68 | 44 | 11 bundles |
| `acdc_lv` | 361 | 17 | 68 | 44 | 10 bundles |
| `riga_cup` | 95 | 17 | 68 | 44 | 9 bundles |
| `riga_disc` | 95 | 17 | 68 | 44 | 11 bundles |
| `brats_wt` | 3228 | 17 | 68 | 44 | 10 bundles / 40 members |

but Table 1 retains the 44-trace primary source subset and 10 bundles after the floor. The authoritative row-level provenance for the table is `within_cross/*.json`, especially the `pool` and `fms_kept_after_floor` fields.

**What else is in the bundle.** In addition to the primary `per_case_dice` tree, the bundle includes released trace sets for decoder capacity, UNet-skip decoding, loss/augmentation sensitivity, fold-reslicing, BraTS train-size, MC dropout, BraTS multimodal input, random-init adapted heads, and a small set of ACDC DINOv2-B full-fine-tuning provenance traces. It also includes selected aggregate diagnostic JSONs under `results/_merged/diagnostics` plus code sufficient to inspect and regenerate the included primary summaries and appendix summaries. Appendix-only diagnostics not listed here, raw prediction arrays, and full pixel-mask caches are not included in the supplement and are treated as reported diagnostics rather than fully bundled reproduction targets.

**What is not in the bundle and why.** Raw medical images, raw labels, feature caches, and full prediction NPZ caches are not included. For restricted datasets such as BraTS, the release is limited to derived per-case Dice and aggregate dependence readouts; users must obtain the raw data from the upstream provider. Larger derived caches and more detailed split manifests are not included in the released artifact.

# NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

Justification: The abstract and introduction claim that *in the frozen and light-adaptation regime of medical segmentation pipelines*, same frozen-checkpoint/recipe pools show higher per-case Dice-score correlation, used as a failure-dependence proxy, than the primary-table cross-bundle reference pools. The primary table uses task-specific functional pools after Dice-floor filtering: Kvasir 11 FMs, ACDC LV 10, BraTS foreground 10, RIGA Cup 9, and RIGA Disc 11. The scope is explicitly restricted to retrospective public-benchmark dependence audits; secondary pixel/lift readouts, released nnU-Net complements, and non-primary mechanism diagnostics are diagnostic or scope context rather than causal clinical-outcome evidence.

2. **Limitations**

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: The main paper gives high-level limitations in Section 8, and the appendix expands them with scoped protocol notes: frozen/light-adaptation only; LoRA and RIGA full-fine-tuning probes without released trace artifacts are retained only as scope notes; released CNN random-init and small ACDC full-fine-tuning provenance traces are diagnostic, not primary evidence; UNet-with-skip and nnU-Net v2 complements reported, including one case-identical RIGA 2D held-out scope check, but not dense 3D FM/full-pipeline baselines for every primary row; the 3D architecture pilot is a scope-boundary diagnostic; 2D axial slices at $224\times224$ for the primary FM audit; MedSAM used only as a frozen vision-encoder probe; broader cross-scanner / cross-demographic shifts remain untested; clinical Dice thresholds are operational conventions and many lift thresholds are support-limited.

3. **Theory assumptions and proofs**

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: The paper is empirical and benchmark-oriented; it does not include theoretical results or proofs. Section 3 introduces estimand notation but the $\bar{r}$, $\Delta$, and lift statistics are descriptive.

4. **Experimental result reproducibility**

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: Section 4, Appendix A13, and Appendix A32 specify the public FM checkpoints, decoder architecture ($1\times1$-conv default plus stated sweeps), training protocol (Adam $10^{-3}$, 30 epochs for the frozen-head primary runs, batch 16, BCE for the binary primary rows), seed set ($\{42, 43, 44, 45\}$ with $\sigma$-seeded determinism), and the actual primary split protocols: Kvasir 880/120, RIGA train = BinRushed+MESSIDOR with Magrabia (Magrabi Eye Center) source held out ($n$=95), ACDC public patients 001–100 with patient-level fold-0 LV-positive slices ($n$=361), and the BraTS 2024 2D FLAIR `seg>0` foreground derivative: per held-out subject, up to the top-10 axial slices by foreground area with at least 64 positive pixels, subject-level 80/20 split using NumPy seed 42 ($n$=3228). This BraTS target includes the GLI resection-cavity label and is not the official 3D BraTS WT challenge target. The supplementary artifact releases primary-task per-case Dice JSONs, merged summaries, selected aggregate diagnostic JSONs, code, command documentation, hashes, and a data card for trace-level reproduction; raw-data retraining requires separately obtaining upstream datasets/checkpoints, and full raw data, full prediction caches, raw held-out prediction arrays, and more detailed split manifests are outside the supplementary artifact.

5. **Open access to data and code**

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: All source datasets used in the reported evidence are public or access-controlled research resources from their upstream providers, subject to their licences, research-use terms, or DUAs: RIGA+, ACDC, Kvasir-SEG, BraTS 2024, MSD tasks, and ISIC-2018. All backbones are publicly available or documented as gated research checkpoints. The supplementary artifact releases benchmark code, estimator implementations, primary-task per-case Dice traces, merged summaries, selected appendix result JSONs, README/MANIFEST commands, hashes, and a data card; authored code and metadata use the bundle licence, while derived scalar traces remain subject to upstream dataset/model terms. Raw medical images, masks, full prediction caches, some held-out-task artifacts, and complete split manifests are not redistributed in the supplementary artifact, so open access is provided for trace-level reproduction rather than raw-data end-to-end retraining.

6. **Experimental setting/details**

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: Section 3 gives the formal estimands (within-/cross-family $\bar{r}$, spatial $\Delta_{\mathrm{pix}}$, threshold lift $L$) and the symmetric $4 \times 4$ seed-combo averaging rule; Section 4 gives the datasets (with patient-level splits), foundation-model list, and protocol (image size, encoder-specific upstream normalisation, decoder, optimiser, loss, epochs, batch, seed control).

7. **Experiment statistical significance**

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: Table 1 reports paired case-level and hierarchical family/seed bootstrap 95% CIs ($B$=2000). The BraTS foreground derivative is analysed at the slice level with an explicit within-subject dependence caveat, Appendix A11 reports subject-level sensitivity checks, and Appendix A12 explains why sparse-class NCR filtering diagnostics are not used as numeric evidence.

8. **Experiments compute resources**

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: Appendix A32 reports the compute/resource scope. For trace-level verification, the artifact is directly reproducible on CPU in minutes through the artifact verification script. Raw-image reproduction used NVIDIA A100-80GB, A6000, and L40S GPUs across a shared cluster. Feature extraction per FM per domain ranges from minutes for CNN backbones to longer ViT/BraTS jobs; cached $1 \times 1$-head training per seed is lightweight. For raw-image retraining, we report resource classes and per-job elapsed times where recorded, but do not claim a single exact total GPU-hour count because raw retraining is outside the submitted artifact's reproduction scope.

9. **Code of ethics**

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

Answer: [Yes]

Justification: The research uses only de-identified public or access-controlled research medical imaging datasets (RIGA, ACDC, Kvasir-SEG, BraTS 2024) under their original research licences, terms, or DUAs. No additional patient data are collected. The paper audits retrospective failure-correlation properties of benchmark segmentation pipelines and does not propose new clinical tools.

10. **Broader impacts**

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: Section 8 discusses deployment implications: shared-pipeline model pools (same FM, same architecture family, same recipe) in frozen/light-adaptation medical AI deployments may be over-estimated as providing independent safety margins. The paper quantifies a retrospective evaluation assumption (independence among FM-pool members) and does not measure clinician decisions, diagnoses, or patient outcomes. Backbone-family diversification is observed to have lower dependence in the tested comparisons, not to be a complete clinical mitigation.

11. **Safeguards**

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [Yes]

Justification: The supplementary artifact does not release raw medical images, voxel labels, full prediction caches, upstream checkpoints, trained clinical services, or patient-facing models. Released artifacts are derived per-case scalar traces, aggregate summaries, metadata, hashes, and scorer code under upstream dataset/model constraints. The paper states that the audit is not a clinical safety guarantee, does not support re-identification or patient profiling, and should not be used as a standalone model-selection or deployment rule.

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13. **New assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

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    Justification: The paper releases (i) a benchmark protocol (estimands + symmetric multi-seed + bootstrap estimators); (ii) primary-task per-case Dice JSONs and merged summary JSONs for the released rows; (iii) selected appendix/ablation result JSONs and reproduction scripts; and (iv) artifact documentation including README/MANIFEST files, SHA-256 hashes, a derived-trace data card, and Croissant metadata submitted for the trace/code artifact. Raw medical images, prediction masks/caches, and full split manifests are not part of the supplementary artifact because upstream redistribution rights and artifact size constraints vary across datasets.

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    Answer: [N/A]

    Justification: No new human-subjects data were collected. Upstream datasets carry their own governance, approvals, or access terms (RIGA, ACDC, Kvasir-SEG, BraTS 2024).

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    Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

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